

DESIGN AND SIMULATION ROADMAP

The GRAIL Collector

Graded Riser Architecture with Integrated Latent-storage

An advanced flat plate solar collector. Design parameters, Fusion 360 modelling, COMSOL 6.2 CFD, MATLAB/Simulink system model, results plan, and patent strategy.

Revision	H — September 2026 · corrected
Scope	B.Tech final year, 30 weeks, simulation only
Tools	Fusion 360 · COMSOL Multiphysics 6.2 · MATLAB/Simulink · Python
Basis	Five supplied FPSC papers plus web and patent prior-art audit
Strongest claim	Alternating counter-current absorber (Filing 1, Section 13.2)

Revision H changes

Revision H corrects Revision G against the CAD model and against itself. Eleven corrections, C1 to C10 plus Decision D4, are applied in place in the sections below and set out in full in the appendix. The most serious, C1, is that the channel grading ran in the wrong direction: A7, 6.1 and 8.2 all specify a converging channel and the model had been built diverging, which put Reynolds number outside the range 8.2 states. All CFD output produced before this correction is void.

One change was a design decision rather than a correction. Decision D4 reduces the root fillet from 1.0 to 0.5 mm, inside the range A11 already permits, because at 1.0 mm the channel section degenerates before the grading of 6.1 can be reached. Section 5.2 also gains two items the built collector has and Revision G omitted, the inner glazing pane and the mounting hardware, and its TOTAL is corrected: the Revision G figure of 17.82 kg was not the sum of its own line items, which came to 22.42 kg.

Contents

Tier system used throughout.....	4
Nomenclature and conventions.....	5
Conventions used throughout.....	5
Latin symbols.....	5
Greek symbols.....	7
Subscripts.....	8
Abbreviations.....	8
Status of every number in this document.....	10
The one physical claim that is proved, and the one that is not.....	10
Verification register.....	11
1. Design thesis.....	13
2. What the five reference papers have not done.....	14
2.1 The fourteen gaps.....	14
2.2 Caution before citing their numbers.....	15
3. Prior-art audit.....	16
4. Complete design parameter set.....	18
4.1 Group A — Absorber channel geometry.....	18
4.2 Group B — Flow-field topology.....	18
4.3 Group C — Optical and surface.....	19
4.4 Group D — Envelope and loss control.....	20
4.5 Group E — Operating conditions.....	20
4.6 Group F — Topology optimisation hyperparameters.....	21
4.7 Derived quantities and objectives.....	21
4.8 Sweep reduction.....	21
5. Collector architecture.....	23
5.1 Layer stack, sun-facing downward.....	23
5.2 Bill of materials.....	23
5.3 Material properties for COMSOL.....	24
6. Fusion 360 modelling.....	25
6.1 Parameter table first.....	25
6.2 Component tree.....	25
6.3 The absorber.....	25
6.4 Deliverables.....	26
7. Fusion 360 to COMSOL 6.2.....	27
7.1 If you must import.....	27
8. CFD methodology in COMSOL 6.2.....	28
8.1 Modelling unit.....	28
8.2 Flow regime.....	29
8.3 Physics interfaces.....	29
8.4 Boundary conditions.....	29
8.5 Mesh.....	30
8.6 Solver.....	30
8.7 Validation.....	30
8.8 Campaign.....	31
9. Topology optimisation.....	32
9.1 COMSOL setup.....	32
9.2 The stagnation constraint.....	32

9.3 Surrogate and multi-objective search.....	32
10. ANN and hybrid ANN modelling.....	34
10.1 Why a surrogate is not optional.....	34
10.2 Dataset preparation.....	34
10.3 Baseline ANN.....	34
10.4 Hybrid ANN models.....	35
10.5 Benchmarks to include.....	37
10.6 Evaluation and acceptance.....	37
10.7 Interpretability.....	37
10.8 Deployment.....	38
10.9 Failure modes and fixes.....	38
11. MATLAB and Simulink system model.....	39
11.1 Why this section exists.....	39
11.2 The CFD-to-Simulink bridge.....	39
11.3 Model architecture.....	39
11.4 Two operating cases.....	40
11.5 Validation of the Simulink model.....	40
12. Results and figures to produce.....	41
12.1 Figures.....	41
12.2 Tables.....	42
12.3 The three headline numbers.....	43
13. Patent strategy.....	44
13.1 Two different things in this document are called patents.....	44
13.2 The three filings from this project.....	44
13.3 Critical terminology warning.....	45
13.4 Third-party patent landscape.....	45
13.5 The two statutory exclusions that govern everything.....	47
13.6 Draft claims.....	47
13.7 Design registration and freedom to operate.....	49
13.8 Search strings to run before filing.....	49
13.9 Costs and timing.....	50
13.10 Action sequence.....	50
14. Schedule and risk.....	52
14.1 Risk register.....	52
14.2 The next seven days.....	53
15. Revision H — corrections and as-built verification.....	54
15.1 Correction register.....	54
15.2 C1 — the grading direction.....	55
15.3 Decision D4 — the root fillet.....	56
15.4 C2 — the grading exponent.....	57
15.5 Two withdrawn claims.....	57
15.6 Note on section 5.3.....	57
15.7 As-built verification record.....	57
15.8 What remains open, and the corrected campaign sequence.....	58
15.9 Cross-check against the draft claims of 13.6.....	59
15.10 C11 — the optical cost of the second glazing pane.....	60
15.11 C12 — numbers that depended on the corrections above.....	61
15.12 C14 — the CFD domain, and a boundary condition that was wrong as specified.....	61
References.....	63
Supplied reference papers.....	63
Prior-art sources.....	63

If empty, open in Word and press Ctrl+A then F9.

Tier system used throughout

Tier	Meaning	Commitment
T1	Core programme. This is the project.	Weeks 4–24, must complete
T2	Stretch, if T1 finishes early.	Weeks 25–28, optional
T3	Designed, deliberately not simulated.	Stated as Future Work

A design novel on nine axes cannot be validated by one student in thirty weeks. Build the whole architecture in CAD; simulate in tiers. A thesis proving two novelties beats one asserting nine.

Nomenclature and conventions

Every symbol used in this document. Symbols are written in the ASCII form used in COMSOL and Fusion 360 parameter tables, with the conventional Greek equivalent given where one exists, so that what you type and what you write in the thesis stay traceable to each other.

Conventions used throughout

Convention	Rule	Example
Parameter names	lowercase with underscores, subsystem prefix first. Type these verbatim into COMSOL and Fusion	ap_L, Dh_in, w_bridge
Greek letters	spelled in ASCII inside code blocks and parameter tables; Greek glyph in running prose	eta in a table, η in prose
Powers of ten	written as 1e-4 in code blocks, and as 10^{-4} in prose and tables	$R_{\text{bond}} = 2\text{e-}4 \text{ m}^2\text{K/W}$
Units	SI throughout. Dimensions in mm unless stated. Temperatures in °C for engineering quantities, K in equations	1100 mm, 130 °C, 300 K
Temperature difference	written ΔT ; always state which two temperatures	$\Delta T = T_{\text{in}} - T_{\text{amb}}$
Currency	Indian rupees, written ₹ or INR	₹1,600
Figure and table IDs	F for figures, T for tables, defined in Section 12	F8, T5
Parameter IDs	letter group plus number, defined in Section 4	A7, B1, C3, F6
Statute references	Section 3(f) and Section 3(k) always mean the Indian Patents Act, never a section of this document	Section 3(f)

Latin symbols

Symbol	Meaning	Unit
A_c	Collector aperture area	m ²
a1	First-order loss coefficient in the efficiency curve	W/m ² ·K
a2	Second-order loss coefficient in the efficiency curve	W/m ² ·K ²
ap_L, ap_W, ap_t	Absorber plate length, width, thickness	mm
b0	Incidence angle modifier coefficient	—
C_add	Embodied carbon of added material	kgCO ₂ e
C_col	Collector thermal capacitance	J/K
ch_AR	Channel aspect ratio, width divided by height	—
ch_pitch	Lateral spacing between channel centrelines	mm
cp	Specific heat capacity at constant pressure	J/kg·K
D, Dh	Channel diameter, hydraulic diameter	m or mm
Dh_in, Dh_out	Hydraulic diameter at inlet and at outlet	mm

Symbol	Meaning	Unit
edge_t	Edge insulation thickness	mm
F_R	Heat removal factor	—
f_interdig	Flag: 0 parallel flow, 1 alternating counter-current	—
f_relief	Emittance relief zone area fraction	—
f_shade	Self-shading fraction of surface features	—
f_topo	Flow field type index, 0 to 3	—
G, G_T	Incident solar irradiance on the collector plane	W/m ²
G_ratio	Axial grading ratio, Dh_out divided by Dh_in	—
gap_t	Air gap between absorber and transparent insulation	mm
h	Convective heat transfer coefficient	W/m ² ·K
J1, J2, J3	Optimisation objective functions	—
k	Thermal conductivity	W/m·K
L	Characteristic length, channel length	m
lam_G	Axial grading profile exponent	—
m_add	Added material mass relative to baseline	kg
mdot	Mass flow rate of working fluid	kg/s
MDI	Manifold maldistribution index, max divided by min channel flow	—
n_alt	Number of channels per flow-direction group	—
n_ch	Number of parallel channels	—
Nu	Nusselt number	—
p	Static pressure	Pa
P_ratio	Pitch grading ratio, outlet pitch divided by inlet pitch	—
pcm_t	Phase change material layer thickness	mm
Pr	Prandtl number	—
q	Heat flux	W/m ²
q_ramp	Brinkman penalisation RAMP parameter	—
Qu	Useful heat gain delivered to the working fluid	W
R4	Radiative loss proxy, area mean of plate temperature to the fourth power	K ⁴
R_bond	Plate-to-channel contact thermal resistance	m ² ·K/W
Re	Reynolds number	—
r_fillet	Channel root fillet radius	mm
r_i	Fin or surface feature interruption ratio	—

Symbol	Meaning	Unit
R_min	Helmholtz filter radius in topology optimisation	mm
T	Temperature	K or °C
T_amb, T_a	Ambient air temperature	K or °C
T_in, T_out	Working fluid temperature at inlet and outlet	K or °C
T_m	Mean working fluid temperature, $(T_{in} + T_{out})/2$	K or °C
T_p	Local absorber plate temperature	K or °C
T_pm	Mean absorber plate temperature	K or °C
T_sky	Effective sky temperature, Swinbank correlation	K
T_stag	Stagnation temperature at zero flow	°C
T_stag_max	Stagnation temperature constraint in optimisation	°C
T_sw	Thermotropic layer switching temperature	°C
tim_t, tim_cell	Transparent insulation thickness, honeycomb cell size	mm
U_L	Overall collector heat loss coefficient	W/m ² ·K
vip_t	Vacuum insulation panel thickness	mm
v_wind	Wind speed over the glazing	m/s
w_bridge	Lateral conduction bridge width between opposed channels	mm
w_therm	Objective weight, thermal versus hydraulic	—
W_p	Pumping power	W
x	Axial coordinate along the flow direction	m or mm

Greek symbols

ASCII form	Greek	Meaning	Unit
alpha_abs	α	Solar absorptance of the absorber surface	—
asym	—	Lateral flow-field asymmetry index	—
beta_tilt	β	Collector tilt from horizontal	deg
beta_proj	β	Projection sharpness in topology optimisation	—
delta_p, dP	Δp	Pressure drop across the collector	Pa
delta_T, dT	ΔT	Temperature difference	K
eps_abs	ϵ	Thermal emittance of the absorber surface	—
eps_glz	ϵ	Thermal emittance of the glazing	—
eps_relief	ϵ	Thermal emittance of the relief zone	—
eta	η	Instantaneous thermal efficiency	—
eta_0	η_0	Optical or zero-loss efficiency	—

ASCII form	Greek	Meaning	Unit
eta_c, eta_m	η	SBX and polynomial mutation distribution indices	—
theta	θ	Solar incidence angle on the collector plane	deg
lam_G	λ	Axial grading profile exponent	—
lambda_1, lambda_2	λ	Loss weighting terms in the physics-informed model	—
mu	μ	Dynamic viscosity	Pa·s
nu	ν	Kinematic viscosity	m ² /s
phi_solid	ϕ	Solid volume fraction in the topology optimisation domain	—
phi_max	ϕ	Maximum solid fraction constraint	—
rho	ρ	Density	kg/m ³
tau_glz	τ	Solar transmittance of the glazing	—
tau_tim	τ	Solar transmittance of the transparent insulation	—
tau_hot	τ	Transmittance above the thermotropic switching point	—
omega	ω	Solar hour angle	deg

Subscripts

Subscript	Meaning	Subscript	Meaning
a, amb	Ambient	in	Inlet
abs	Absorber	m	Mean
add	Added relative to baseline	max, min	Maximum, minimum
c	Collector	out	Outlet
ch	Channel	p	Plate
exp	Experimental	pm	Plate mean
glz	Glazing	stag	Stagnation
hot	Above switching temperature	tim	Transparent insulation

Abbreviations

Abbrev.	Meaning	Abbrev.	Meaning
AI	Artificial intelligence	MAPE	Mean absolute percentage error
ANFIS	Adaptive neuro-fuzzy inference system	MLP	Multilayer perceptron
ANN	Artificial neural network	MMA	Method of moving asymptotes
AR	Anti-reflective	MSE	Mean squared error
BOM	Bill of materials	NSGA-II	Non-dominated sorting genetic algorithm II
CFD	Computational fluid dynamics	PC	Polycarbonate

Abbrev.	Meaning	Abbrev.	Meaning
CPC	Cooperative Patent Classification	PCM	Phase change material
DHW	Domestic hot water	PINN	Physics-informed neural network
EPO	European Patent Office	PSO	Particle swarm optimisation
FPSC	Flat plate solar collector	PVT	Photovoltaic-thermal
FTO	Freedom to operate	RMSE	Root mean square error
GA	Genetic algorithm	SBX	Simulated binary crossover
GD&T	Geometric dimensioning and tolerancing	SHAP	SHapley Additive exPlanations
GPR	Gaussian process regression	SMA	Shape memory alloy
HWB	Hottel-Whillier-Bliss	TIM	Transparent insulation material
IAM	Incidence angle modifier	TMY	Typical meteorological year
InPASS	Indian Patent Advanced Search System	TO	Topology optimisation
IPR	Intellectual property rights	TPMS	Triply periodic minimal surface
KAN	Kolmogorov-Arnold network	UDF	User-defined function
LHS	Latin hypercube sampling	VIP	Vacuum insulation panel
MAE	Mean absolute error	XGBoost	Extreme gradient boosting

Status of every number in this document

Read this before quoting any figure

Not one performance number in this document is a simulation result, because the simulations have not been run yet. The efficiency figures, the loss coefficients and the stagnation temperatures are DESIGN TARGETS computed from standard correlations and typical component properties. They tell you what to aim at and what to test. They are not findings and must never be presented as such in a thesis, a paper or a patent specification.

The table below classifies every class of quantity in this document by where it came from. Check this table before you quote anything.

Class	Meaning	Where it appears	Before you rely on it
DESIGN TARGET	Computed from textbook correlations and typical component properties. Not simulated, not measured.	Section 1 efficiency table; U_L 2.1 W/m ² K; eta_0 0.66; a1 2.00; +27 % at 40 K; stagnation ≈135 °C; the 130 °C constraint	Replace with your own CFD output. Until then, cite as a design objective
TYPICAL	Manufacturer-typical or handbook values for the material class, not a specific product datasheet.	Section 5.3 material properties; TIM conductivity and transmittance; PCM latent heat; VIP conductivity	Confirm against the actual supplier datasheet before procurement and before final simulation runs
INDICATIVE	Order-of-magnitude cost estimates for Indian single-unit sourcing, not quotations.	Section 5.2 bill of materials; the ₹51,100 total	Obtain quotations if cost forms part of your analysis
LITERATURE	Quoted directly from the five reference papers or from other published work.	Section 2 claimed gains; validation targets; Sakib's R ² values; the eight to ten per cent numerical-experimental discrepancies	Verify against the source paper before quoting. Note the internal inconsistencies listed in Section 2.2
STATUTORY	Official published fee schedules and statute text.	Section 13.9 patent fees; Sections 3(f), 3(k) and 31 of the Patents Act; the Designs Act term	Confirm current fees on the Indian Patent Office site at the time of filing
SEARCH RESULT	Located by a web search in September 2026. Databases of record were NOT searched.	Every OPEN and every Nothing located entry in Sections 3 and 13.4	Re-run on Scopus, Web of Science, Espacenet and InPASS before asserting novelty anywhere. This is the single largest unclosed risk in the project
SETTING	Recommended software configuration values.	Mesh counts, solver tolerances, ANN hyperparameters, NSGA-II settings, TO hyperparameters	Starting points, not optima. Expect to tune them

The one physical claim that is proved, and the one that is not

The Jensen's inequality argument in Section 4.2 is mathematically exact: for a fixed mean plate temperature, a flatter temperature field has a strictly lower area mean of the fourth power of temperature, so it radiates less. That direction is guaranteed and cannot fail.

What is entirely unknown is the magnitude. The alternating counter-current arrangement might reduce the radiative loss proxy R4 by eight per cent, in which case the effect is real, patentable and worth building a thesis on. It might reduce it by half a per cent, in which case it is a curiosity. Nothing in this document tells you which, and no amount of further reading will. Figures F8 to F11 decide it.

Consequence for the project plan

Run the counter-current comparison early — a three-channel strip with f_{interdig} set to 0 and then 1, at matched mean plate temperature and matched mass flow, reporting R4. It is the first thing that should come out of the week 13 to 18 campaign, not the last.

If R4 falls by a useful margin, the patent filing at week 22 is justified and the thesis has its headline. If it does not, you have lost one week rather than one term, and the topology-optimisation and grading contributions still stand on their own.

Verification register

Every item below must be closed before the corresponding output is presented as a result. Track it as a live checklist.

#	What to verify	How	By when	Blocks
V1	Prior-art null results for Filings 1, 2 and 3	Scopus, Web of Science, Espacenet, InPASS using the Section 13.8 strings. Log every string, database, date and hit count	Week 6	Patent filing, novelty claims
V2	COMSOL licence covers CFD, Heat Transfer, Optimization and CAD Import modules	Check the licence file or ask your administrator	Week 1	Sections 8, 9
V3	MATLAB licence covers Simulink; whether Simscape and Stateflow are included	Check the licence	Week 1	Section 11
V4	COMSOL 6.2 feature names and menu paths in Sections 8 and 9	Open the software and confirm each node exists where stated. Version differences are common	Week 4	Model build
V5	Baseline model validation below 8 % against published data	Section 8.7 procedure	Week 8	Everything downstream
V6	Material properties against supplier datasheets	Request datasheets for TIM, VIP, PCM and selective coating	Week 9	Final simulation runs, BOM
V7	Magnitude of the counter-current effect on R4	Three-channel strip, f_{interdig} 0 vs 1, matched mean plate temperature	Week 15	Filing 1
V8	Efficiency coefficients η_{a0} , a_1 , a_2 from your own CFD	Fit to at least 12 CFD points per Section 11.2	Week 24	Section 11, all annual results
V9	Interpreted topology-optimised geometry re-simulated	Export, smooth, re-mesh, re-run conjugate heat transfer	Week 25	TO performance claims
V10	Claim language reviewed by a registered patent agent	Via your institution's IPR cell	Week 21	Patent filing

#	What to verify	How	By when	Blocks
V11	Current official patent and design fees	Indian Patent Office website at time of filing	Week 21	Budget
V12	All CFD output re-run on the corrected converging geometry	Confirm on the built model that $Dh_{out} = Dh_{in} * G_{ratio}$ and that Re falls in 23.9 to 198.8 BEFORE any run. Discard every CFD result produced before Revision H: they were computed on a diverging channel and are void, not approximate	Week 8	Everything downstream, Filings 1, 2 and 3

1. Design thesis

Every collector in the reference set improves the same term. This one improves the other.

$$\eta = \frac{F_R(\tau \cdot \alpha)}{|\text{optical}|} - \frac{F_R U_L (T_i - T_a)}{G} \quad \text{thermal loss}$$

Freegah's elliptical risers, Theeyzen's wire mesh, Khanjian's wavy fins and Shivanayak's coil inserts all raise F_R . None touches U_L . Yet at 40 K temperature difference under 800 W/m², the loss term destroys about 43 % of the energy already captured. The field is optimising the smaller term.

GRAIL attacks U_L with transparent insulation and vacuum backing, accepts an optical penalty of roughly 25 per cent for the double-glazed stack as built — 18 per cent with a single pane, see correction C11 — and recovers F_R by eliminating the plate-to-tube bond with a monolithic roll-bonded channel plate.

Every figure in the table below is a DESIGN TARGET computed from standard correlations and typical component properties, not a simulation result. See Status of every number. Replace them with your own CFD output before presenting any of them as a finding.

Note added in Revision H. The optical row above assumes a single glazing pane. The built collector has two, so the correct optical term is $\tau_{\text{glz_sys}}$ 0.833 rather than 0.91, which takes $F_R(\tau \cdot \alpha)$ from 0.66 to about 0.60. The efficiency rows have NOT been recomputed, and deliberately so: the second pane also lowers U_L by splitting the cavity, and the $F_R \cdot U_L$ figure of 2.00 in this table was computed for the single-pane stack. One term moves down and the other moves down too, so the net effect on efficiency is genuinely undetermined until both are computed together. What can be said now is that the crossover against a conventional collector, shown here at 20 K, moves later if only the optical term is corrected — to roughly 30 K — and moves back toward 20 K as the U_L gain is credited. Treat every efficiency figure in this table as a single-glazed design target until the comparison in 15.10 is run.

Parameter	Conventional	GRAIL	Basis
$F_R(\tau \cdot \alpha)$	0.78	0.66	$\tau_{\text{glz}} 0.91 \times \tau_{\text{TIM}} 0.82 \times \alpha 0.95$ — SINGLE PANE. As built the collector is double glazed: see the note below and correction C11
$F_R \cdot U_L$ [W/m ² K]	6.75	2.00	TIM front + VIP rear
F_R	0.90	0.93	no bond resistance
η at $\Delta T = 20$ K	0.61	0.61	crossover
η at $\Delta T = 40$ K	0.44	0.56	+27 % relative
η at $\Delta T = 60$ K	0.27	0.51	+89 % relative

The argument that matters, and it is not the percentage

Conventional flat plates collapse above about 70 °C — exactly where process heat, sterilisation and pasteurisation live, so those loads default to fossil fuel or evacuated tubes at three times the cost. A flat plate holding useful efficiency to 90 °C opens an application band flat plates cannot presently serve. Prove this with the Simulink annual model of Section 11, not with a single-point efficiency.

2. What the five reference papers have not done

#	Reference	Enhancement	Claimed gain
P2	Freegah et al., J. Eng. Res. 13 (2025) 898	Elliptical riser, channels, wavy fins	+22.4 %
P3	Theeyzen & Freegah, Results in Eng. 24 (2024) 103203	Wire mesh on riser and plate	+41.5 %
P11	Shivanayak et al., Int. J. Thermofluids 29 (2025) 101378	Wavy riser tubes, coil inserts	eta = 84 %
P13	Sakib et al., Energy Convers. Manag. X 31 (2026) 102073	None — KAN surrogate + NSGA-II	R ² = 0.992
P15	Khanjian et al., Results in Eng. 30 (2026) 111196	Longitudinal vs wavy plate fins	+13.2 %

Papers 2, 3 and 15 come from one laboratory and share a rig, an equation set and an uncertainty method; paper 15 reuses paper 2's validation errors and baseline. Paper 13 validates against paper 2. Their shared blind spots are therefore systematic, not coincidental.

2.1 The fourteen gaps

#	What is missing	Why it matters	Addressed by
1	No paper addresses U_L	All improve F_R only, leaving the larger loss term untouched	TIM, VIP
2	Bond resistance never modelled	0.5e-4 to 2e-4 m ² K/W is the largest unforced loss; omitting it is why simulated collectors beat built ones	Monolithic plate, A12
3	Stagnation never mentioned	Flow stoppage drives absorbers to 180–220 °C, destroying fluid and seals	F6, Construction 4
4	Flow distribution never simulated	All model a single tube; real headers give 1.3–1.6 maldistribution	B7
5	No enhancement varies along the flow	Driving temperature difference collapses downstream, yet enhancement is uniform	A7–A9
6	No formal geometry optimisation	Three or four discrete cases, then declare a winner	Sections 9, 10
7	Topology optimisation never used	The flow field is always a guess, never a computed optimum	Section 9
8	Pressure drop ignored in three of five	Papers 2, 3 and 15 report no hydraulic penalty, so gains are unpriced	Objective J2
9	Material cost never reported	Covering an absorber in mesh is a lot of copper; nobody counts it	Objective J3
10	Enhanced geometries rarely validated	Papers 3 and 15 validate only the baseline	Section 8.7
11	Single-day performance only	No annual yield, no part-load, no off-design	Section 11, Simulink

#	What is missing	Why it matters	Addressed by
12	Soiling ignored	Indian sites lose 15–30 %, dwarfing most reported gains	T3
13	End-of-life and embodied carbon ignored	Laminated collectors are landfilled	Objective J3
14	P13 optimises, but on a plain single tube	Excellent framework, trivial geometry — no fins, no topology, steady state	Sections 8–10

2.2 Caution before citing their numbers

The claimed gains are not comparable. Paper 3 uses a 1.5 m riser and 15.6 L tank; papers 2 and 15 use 1 m and 9.6 L. Baselines, initial temperatures and test days all differ. There are also internal inconsistencies:

- P15 swaps Model I/II and A/B mid-text; its conclusion states 5.17 % and 13.21 % against body figures of 64.4 % and 68.8 % versus a 55.1 % baseline, which are 9.3 and 13.7 points. The numbers do not reconcile.
- P3's mesh table lists Models D and E that appear nowhere else; its 1 p.m. tank temperatures place Model C, the best model, lowest. Abstract says 41.4 %, conclusion 41.5 %.
- P11 reports 84 % (abstract, 10 mm) against 82 % (text, 30 mm); its CFD is explicitly limited to flow patterns, so efficiency claims are experimental only.

Cite them respectfully, but normalise onto a common baseline and state why published percentages cannot be ranked directly. That observation is itself a contribution.

3. Prior-art audit

The branching manifold is already patented

A Murray's-law biomimetic branching structure in a roll-bonded aluminium absorber is FracTherm, developed by Dr Michael Hermann at Fraunhofer ISE under the EU BIONICOL project (2008–2011), winner of the 2008 International Bionic Award, patented by Fraunhofer and licensed to CGA Technologies S.p.A. in June 2013 specifically for roll-bond solar absorbers.

This is exactly the proposed subsystem in exactly the proposed process. It cannot be claimed and cannot be patented. Drop it: Construction 1 does not require branching, so removing the manifold also removes the licensing issue entirely.

Subsystem	Prior art	Status
Branching manifold in roll-bond	FracTherm, Fraunhofer ISE, patented, licensed 2013	CLAIMED
Roll-bond vs sheet-and-tube	Energy (2013); standard in commercial PVT	CLAIMED
Thermotropic overheat layers	Sol. Energy Mater. (2008); Solar Energy (2009); ASME JSEE 137 (2015) 021003	CLAIMED
Switchable / thermochromic emittance	EPFL switchable selective absorbers; VO ₂ :Ge, Sol. Energy Mater. Sol. Cells (2022)	CLAIMED
SMA / bimetallic stagnation vent	Renewable Energy (1998); PT1592929E; ES2650290A1; WO1981000613A1	CLAIMED
PCM behind the absorber	Heavily published 2020–2025	CLAIMED
TIM, honeycomb, aerogel glazing	Dedicated review: front-side heat loss reduction, Energy Reports (2023)	CLAIMED
Vacuum glazing in collectors	Int. J. Photoenergy (2017); Solar Energy (2021); DE10037088; US10889501B2	CLAIMED
Variable cross-section absorber	Springer JEAS (2025) — but varies PLATE THICKNESS LATERALLY, not channel diameter axially	ADJACENT
Bionic leaf-vein collector	Appl. Therm. Eng. (2025) — PVT systems	CLAIMED for PVT
Topology optimisation of heat sinks	Extensive: variable density, pseudo-3D, 3D microchannel	CLAIMED for heat sinks
Topology optimisation of a solar receiver	Entropy 25 (2023) 398 — parabolic dish cavity only	CLAIMED for dish
Spiral alternating-flow polymer collector	Applied Sciences 14 (2024) 11045 — spiral, polymer, no homogenisation rationale	ADJACENT
TO of a FLAT PLATE absorber	Nothing located	OPEN
Axial channel diameter + pitch grading	Nothing located	OPEN
Alternating counter-current channels	Nothing located in a metal flat plate absorber	OPEN
Static spatially patterned emittance zones	Nothing located	OPEN, narrow

Stagnation-constrained optimisation	Nothing located	OPEN
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Nothing located in a web search is not proof of absence. Before asserting novelty, run the Section 13.8 search strings on Scopus, Espacenet, Google Patents and InPASS, and record every string, database and date in an appendix. A documented search is evidence; an undocumented claim is not.

4. Complete design parameter set

Use these symbols verbatim in both Fusion 360 User Parameters and COMSOL Global Parameters, so the two models stay traceable. Role: SWEPT varied in the campaign; FIXED held constant; DERIVED computed from others. Novelty: NEW nothing located; ADJ adjacent prior art, narrow the claim; EST established.

4.1 Group A — Absorber channel geometry

ID	Symbol	Parameter	Unit	Baseline	Range	Role	Nov.
A1	ap_L	Plate length	mm	1100	750–1250	SWEPT	EST
A2	ap_W	Plate width	mm	480	fixed	FIXED	EST
A3	ap_t	Plate thickness	mm	2.0	1.5–4.0	SWEPT	EST
A4	n_ch	Number of channels	—	12	8–16	SWEPT	EST
A5	ch_pitch	Channel pitch	mm	40	ap_W/n_ch	DERIVED	EST
A6	Dh_in	Hydraulic dia. at inlet	mm	5.2	3.5–7.0	SWEPT	EST
A7	G_ratio	Grading ratio Dh_out/Dh_in	—	1.00	0.35–1.50	SWEPT	NEW
A8	lam_G	Grading profile exponent	—	1.0	0.5–3.0	SWEPT	NEW
A9	P_ratio	Pitch grading ratio	—	1.00	0.60–1.40	SWEPT	NEW
A10	ch_AR	Channel aspect ratio w/h	—	2.1	1.0–4.0	SWEPT	EST
A11	r_fillet	Channel root fillet	mm	0.5	0.5–2.0	SWEPT	EST
A12	R_bond	Contact resistance	m ² K/W	0	0–2e-4	SWEPT	NEW

$$Dh(x) = Dh_{in} * [1 - (1 - G_{ratio}) * (x/ap_L)^{lam_G}]$$

lam_G < 1 concave (taper early) lam_G = 1 linear
 lam_G > 1 convex (taper late) G_ratio = 1 no grading

A12 matters: no reference paper models the bond at all, so the benefit of eliminating it has never been quantified. Sweeping it separates the gain from manufacturing route versus geometry, and pre-empts the obvious reviewer question.

Note on A7 added in Revision H: dependent claim 1 of Filing 1 in 13.6 caps the grading ratio at 1.00, narrower than the 0.35 to 1.50 given here, and requires hydraulic diameter to decrease monotonically. Sweep points above 1.00 are diverging channels outside that claim. They may be studied, but nothing drawn from them may be recited in support of Filing 1. See 15.9.

4.2 Group B — Flow-field topology

The strongest novelty in the project. B1–B3 define the alternating counter-current arrangement.

ID	Symbol	Parameter	Unit	Baseline	Range	Role	Nov.
B1	f_interdig	Alternating counter-current: 0 parallel, 1 alternating	—	0	0/1	SWEPT	NEW

ID	Symbol	Parameter	Unit	Baseline	Range	Role	Nov.
B2	n_alt	Channels per direction group	—	1	1–4	SWEPT	NEW
B3	w_bridge	Lateral conduction bridge width	mm	34.8	20–60	SWEPT	NEW
B4	f_topo	Field type: 0 straight, 1 graded, 2 serpentine, 3 TO	—	0	0–3	SWEPT	NEW
B5	phi_solid	Solid volume fraction, TO domain	—	—	0.15–0.40	SWEPT	NEW
B6	asym	Lateral asymmetry index	—	0	0–0.30	SWEPT	NEW
B7	MDI	Maldistribution index max/min	—	1.00	report	DERIVED	EST

Why B1–B3 matter more than they look

Radiative loss scales as the fourth power of surface temperature. By Jensen's inequality, at fixed mean plate temperature a flatter field has a strictly lower mean of the fourth power of temperature. Alternating the flow direction so hot outlet regions sit laterally adjacent to cold inlet regions homogenises the plate by conduction.

You therefore reduce total loss without improving heat transfer at all — you rearrange where the heat is. That is provable, and it is the opposite of what every paper in the reference set does.

B3 controls it: wide enough to carry lateral conduction, narrow enough that the streams do not thermally short-circuit. Expect an interior optimum, which is what makes it worth optimising.

4.3 Group C — Optical and surface

ID	Symbol	Parameter	Unit	Baseline	Range	Role	Nov.
C1	alpha_abs	Solar absorptance	—	0.95	0.90–0.96	FIXED	EST
C2	eps_abs	Emittance, main zone	—	0.04	0.03–0.10	FIXED	EST
C3	eps_relief	Emittance, relief zone	—	0.04	0.04–0.45	SWEPT	ADJ
C4	f_relief	Relief zone area fraction	—	0	0–0.25	SWEPT	ADJ
C5	tau_glz	Glazing transmittance	—	0.91	0.88–0.92	FIXED	EST
C6	eps_glz	Glazing emittance	—	0.88	fixed	FIXED	EST
C7	f_shade	Self-shading fraction	—	0	0–0.15	DERIVED	NEW
C8	r_i	Feature interruption ratio	—	0	0–0.40	SWEPT	NEW
C9	tau_glz_sys	Glazing SYSTEM transmittance, all panes	—	0.83	0.68–0.92	DERIVED	EST

C3 and C4 are ADJ, not NEW: switchable and thermochromic coatings are established. What was not located is a static, spatially patterned emittance gradient with no switching material. Any claim must recite spatial patterning and expressly disclaim thermochromic materials.

C7 is unclaimed and nobody computes it. Theeyzen argues qualitatively that mesh shades less than fins but never calculates the shadow. Define C7 as projected feature shadow area over aperture area at the design incidence angle and subtract it from absorbed flux. This will change the ranking of finned designs.

4.4 Group D — Envelope and loss control

ID	Symbol	Parameter	Unit	Baseline	Range	Role	Nov.
D1	tim_t	TIM thickness	mm	0	0–20	SWEPT	EST
D2	tim_cell	Honeycomb cell size	mm	3.5	2.0–6.0	SWEPT	ADJ
D3	k_tim	TIM through-thickness k	W/mK	0.075	0.018–0.12	DERIVED	EST
D4	tau_tim	TIM transmittance	—	0.82	0.62–0.88	DERIVED	EST
D5	gap_t	Absorber-to-TIM gap	mm	15	8–30	SWEPT	EST
D6	vip_t	VIP thickness	mm	20	10–30	SWEPT	EST
D7	k_vip	VIP core k	W/mK	0.006	0.006–0.010	FIXED	EST
D8	edge_t	Edge insulation	mm	15	10–25	FIXED	EST
D9	T_sw	Thermotropic switch temp.	°C	—	70–85	SWEPT	EST
D10	tau_hot	Transmittance when switched	—	—	0.30–0.50	SWEPT	EST
D11	glz_n_panes	Number of glazing panes	—	2	1–2	SWEPT	EST

D2 is ADJ because honeycomb TIM is established, but co-optimising cell geometry jointly with the absorber flow field is not — front-side insulation and absorber geometry are always optimised separately. Treating D2 and Groups A–B as one coupled design space is a defensible methodological first.

4.5 Group E — Operating conditions

Site and duty conditions, not design choices. Sweep to train the surrogate; hold FIXED during optimisation, because optimising over conditions a designer cannot control produces a meaningless best design.

ID	Symbol	Parameter	Unit	Baseline	Range	Campaign	In optim.
E1	G_T	Irradiance	W/m ²	800	400–1000	SWEPT	FIXED 800
E2	T_in	Inlet temperature	K	300	285–330	SWEPT	FIXED 300
E3	T_amb	Ambient temperature	K	298	288–318	SWEPT	FIXED 298
E4	mdot	Total mass flow	kg/s	0.0025	0.001–0.0045	SWEPT	SWEPT
E5	v_wind	Wind speed	m/s	1.0	0–5	SWEPT	FIXED 1.0
E6	beta_tilt	Tilt	deg	33	10–45	FIXED	FIXED

4.6 Group F — Topology optimisation hyperparameters

ID	Symbol	Parameter	Baseline	Range	Note
F1	R_min	Helmholtz filter radius	3.0 mm	2.0–6.0	Must exceed 2.5× local mesh size AND the roll-bond minimum feature. Decides manufacturable versus fractal
F2	beta_proj	Projection sharpness	staged	1→2→4→8→16	Never start high; use each solution as the next initial guess
F3	q_ramp	Brinkman RAMP parameter	0.01	0.001–0.1	How hard solid resists flow
F4	phi_max	Max solid fraction	0.40	0.15–0.50	Without it the optimiser fills the domain
F5	w_therm	Thermal vs hydraulic weight	0.7	0.3–0.9	Sweep to trace an internal trade-off curve
F6	T_stag_max	Stagnation constraint	130 °C	110–180	The parameter that makes the study original

4.7 Derived quantities and objectives

Useful heat gain	Q_u	$= \dot{m} \cdot c_p \cdot (T_{out} - T_{in})$	[W]
Thermal efficiency	η	$= Q_u / (A_c \cdot G_T)$	[-]
Pressure drop	dP	$= p_{in} - p_{out}$	[Pa]
Pumping power	W_p	$= dP \cdot \dot{m} / \rho$	[W]
Loss coefficient	U_L	$= (q_{abs} - Q_u / A_c) / (T_{pm} - T_{amb})$	[W/m ² K]
Plate spread	dT_p	$= T_{p,max} - T_{p,min}$	[K]
Radiative proxy	R_4	$= \text{mean}(T_p^4)$ over absorber surface	[K ⁴]
Stagnation temp.	T_{stag}	solve with $\dot{m} \rightarrow 0$, full irradiance	[C]
Added mass	m_{add}	$= \rho_{Al} \cdot (V_{design} - V_{baseline})$	[kg]
Embodied carbon	C_{add}	$= m_{add} \cdot 0.6$ (recycled Al)	[kgCO ₂ e]

R_4 is the quantity the alternating counter-current field is designed to minimise and the metric that demonstrates the mechanism. Two designs with identical mean plate temperature but different R_4 will have measurably different losses — that comparison is the proof.

Objective	Direction	Expression	Use
J1 thermal	maximise	Q_u , or η at design point	Always
J2 hydraulic	minimise	dP , or W_p	Always
J3 material	minimise	m_{add} or C_{add}	Optional third
C1 stagnation	constraint	$T_{stag} \leq T_{stag_max}$	The original contribution
C2 manufacturability	constraint	$\text{min feature} \geq R_{min}$	Enforced by filter
C3 flow balance	constraint	$MDI \leq 1.10$	Full-panel models only

4.8 Sweep reduction

Twenty-three parameters are marked SWEPT, one more than Revision G because Decision D4 moved r_{fillet} from FIXED to SWEPT. That is too many for 250 runs. Fix all of Group D except tim_t and vip_t ; fix α_{abs} ,

eps_abs, tau_glz, eps_glz, v_wind, beta_tilt. Revision H no longer fixes r_fillet: Decision D4 made it a swept parameter, because it sets the floor on how far the channel can be graded and it shares the lateral conduction path with w_bridge. That leaves about sixteen, which 250–350 LHS samples still support. If the surrogate underperforms, drop ap_t and R_bond next — Sakib found plate thickness statistically insignificant for both outputs.

Recommended active set: ap_L, n_ch, Dh_in, G_ratio, lam_G, P_ratio, ap_t, f_interdig, n_alt, w_bridge, r_fillet, f_relief, r_i, G_T, T_in, mdot.

5. Collector architecture

Property	Value	Property	Value
Gross	1200 × 600 × 92 mm	Aperture	0.55 m ²
Absorber	1100 × 480 mm	Tilt	33°
Dry mass	17.8 kg	Working fluid	Water
Flow regime	Laminar, Re 30–210/channel	Target U _L	2.1 W/m ² K

5.1 Layer stack, sun-facing downward

Layer	Material	Thick.	Tier
Self-cleaning + AR coating	TiO ₂ sol-gel over porous SiO ₂	~150 nm	T3
Outer glazing	Low-iron tempered glass, tau 0.91	3.2 mm	T1
Thermotropic overheat layer	PMMA blend, T _{sw} 75 °C	1.0 mm	T2
Transparent insulation	PC capillary honeycomb, 3.5 mm cell	10 mm	T1
Sealed air gap	Dry air, PTFE breather vented. Divided by the inner pane into 7.45 mm lower and 4.30 mm upper	15 mm envelope	T1
Inner glazing pane	Low-iron tempered glass, second pane	3.2 mm	T1
Selective coating	Sputtered TiNOX, alpha 0.95 / eps 0.04	~0.3 um	T1
Graded microchannel absorber	Roll-bonded AA1050, 2 × 1.0 mm	2.0 mm	T1
Thermal interface	Compressed graphite sheet	0.4 mm	T2
PCM latent buffer	RT55 + 10 wt% expanded graphite	10 mm	T2
Vacuum insulation panel	Fumed silica, k 0.006 W/mK	20 mm	T1
Backsheet	Recycled aluminium, powder coated	1.0 mm	T1
Frame	Recycled Al 6063-T6, snap-fit	92 mm o/a	T3

5.2 Bill of materials

Indicative Indian single-unit sourcing, 2026.

Item	Specification	kg	INR
Outer glazing	Low-iron tempered, 1160 × 560 × 3.2 mm, AR + TiO ₂	5.20	2,800
Inner glazing pane	Low-iron tempered, 3.2 mm, second pane	4.294	TBD
Thermotropic film	PMMA, T _{sw} 75 °C, 1.0 mm, EVA laminated	0.78	6,500
Transparent insulation	PC honeycomb, 3.5 mm cell × 10 mm	0.42	3,200
Absorber plate	Roll-bond AA1050, 1100 × 480 × 2.0 mm	2.85	9,000
Selective coating	TiNOX sputtered	—	2,400
Manifold blocks	CNC aluminium, 2 off	0.35	4,500

Item	Specification	kg	INR
Graphite interface	Compressed natural graphite, 0.4 mm	0.21	1,100
PCM	RT55 + 10 wt% expanded graphite	4.65	5,600
PCM tray	Al 0.8 mm, ribbed, nitrogen filled	1.30	2,200
VIP	Fumed silica, 1100 × 500 × 20 mm	2.09	4,800
Edge insulation	Aerogel blanket, 15 mm	0.12	1,400
Frame extrusion	Recycled Al 6063-T6, 3.6 m	2.90	2,600
Backsheet	Recycled Al 1.0 mm, powder coated	1.95	900
Seals and fasteners	EPDM, butyl, PTFE breather, SS316 M5	0.44	1,200
Sensing package	6 × DS18B20, ESP32, Si pyranometer	0.20	2,900
Mounting hardware	Tilt rails, brackets, pivot pins, lock bolts	6.378	TBD
TOTAL		34.13	51,100

5.3 Material properties for COMSOL

Material	rho kg/m ³	cp J/kgK	k W/mK	Other
Water at 300 K	997	4180	0.610	mu 8.55e-4 Pa·s
Aluminium AA1050	2705	900	229	—
Aluminium 6063-T6	2700	900	201	—
Low-iron solar glass	2500	750	1.05	tau 0.91, eps 0.88
PC honeycomb, homogenised	85	1200	0.075 perp / 0.16 para	tau 0.82
Silica aerogel, granular	120	1000	0.020	tau 0.62
TiNOX coating	—	—	boundary condition	alpha 0.95, eps 0.04
RT55 + 10 % EG, solid	880	2000	2.80	L 170 kJ/kg
RT55 + 10 % EG, liquid	770	2200	2.40	T _m 51–57 °C
Graphite interface	1100	710	300 para / 5 perp	—
VIP fumed silica	190	850	0.006	derate 20 % / 25 y
Aerogel blanket	150	1000	0.015	—
EPDM gasket	1150	2000	0.25	—

6. Fusion 360 modelling

6.1 Parameter table first

Modify > Change Parameters > User Parameters. Create all of these before opening a sketch. Skip this and you will rebuild the model three times. Use lowercase with underscores; Fusion accepts spaces then silently fails to resolve the expression.

Parameter	Expression	Meaning
ap_L / ap_W / ap_t	1100 / 480 / 2.0 mm	Absorber plate
n_ch	12	Channel count
ch_pitch	ap_W / n_ch	Driven, never typed
Dh_in	5.2 mm	Inlet hydraulic diameter
G_ratio	0.54	Grading ratio
Dh_out	Dh_in * G_ratio	Driven
ch_w_in / ch_h_in	8.700954 / 4.132953 mm	Lenticular section at inlet, re-solved so Dh_in = 5.2 holds with r_fillet 0.5
r_fillet	0.5 mm	Channel root fillet. Decision D4, was A11 baseline 1.0
lam_G	1.0	Grading profile exponent, A8 baseline
coat_t	0.0003 mm	Selective coating. Bind the coating offset to this parameter
glz_inner_t	3.2 mm	Inner glazing pane, second pane
vip_L	1100 mm	VIP length, trimmed from 1120 mm to clear the manifolds
glz_t / tim_t / gap_t	3.2 / 10 / 15 mm	Front stack
pcm_t / vip_t	10 / 20 mm	Rear stack
frame_d / tilt	92 mm / 33 deg	Envelope and assembly

6.2 Component tree

Create as separate components in one assembly, not bodies in one component. This is what lets you suppress subsystems for the simulation export.

00_Frame · 01_Glazing · 02_TIM · 03_Absorber · 04_Manifold · 05_PCM_Tray
· 06_Insulation · 07_Backsheet · 08_Seals · 09_Sensing

6.3 The absorber

1. Sketch the channel centreline on the plate midplane; fully constrain it.
2. Create profile sketches perpendicular to the centreline at both ends: lenticular ch_w_in by ch_h_in at inlet, and at outlet the same section scaled so that the resulting hydraulic diameter satisfies Dh_out = Dh_in * G_ratio. Note that the section SCALE is not equal to G_ratio, because the root fillet is fixed in size and does not scale with the section. In the built model a section scale of 0.527853 produces a Dh ratio of 0.540339 against the 0.54 that 6.1 states. Solve the scale from the target hydraulic diameter; never type G_ratio in as a scale factor.

3. Loft between them using the centreline as a rail, in Rails mode. Changing G_{ratio} now regenerates the channel.
4. For non-linear grading, insert intermediate sections at $ap_L/3$ and $2 \cdot ap_L/3$ with widths from the grading expression. Three sections cover linear, convex and concave from one feature.
5. Rectangular Pattern the channel, n_{ch} instances at ch_{pitch} .
6. Extrude the plate, then Combine > Cut with the channel bodies, keeping tools. You now have the solid plate and a separate fluid-volume body — exactly what you need later.
7. Add the root fillet where the channel meets the flat sheet, at r_{fillet} , which Revision H sets to 0.5 mm under Decision D4. That fillet is where the real conduction path lives; a sharp corner exaggerates results. Do not hard-code it — drive it from r_{fillet} , because A11 now sweeps it and because the section geometry degenerates if it grows relative to the channel height.
8. For the alternating counter-current arrangement, mirror alternate channels end-for-end and route them to the opposite manifold. Keep w_{bridge} as a driven parameter.

Do not model the honeycomb cells

A 10 mm honeycomb at 3.5 mm cell across 0.55 m^2 is about 45,000 cells. It will make Fusion unusable and COMSOL impossible. Model the TIM as one solid body with an appearance override, and build the cell structure only in a $40 \times 40 \text{ mm}$ cutaway block for renders.

6.4 Deliverables

- Joints, not rigid groups, so you can produce an exploded view (Animation > Auto Explode All Levels).
- Section view through the layer stack at mid-length — your single most useful thesis figure.
- 2D drawing of the absorber with the grading table and GD&T on plate flatness; roll-bond plates bow.
- Two renders: complete and exploded, physically based aluminium and glass.

7. Fusion 360 to COMSOL 6.2

There is no LiveLink for Fusion 360

COMSOL supplies LiveLink for SOLIDWORKS, Inventor, Creo, Solid Edge, AutoCAD and Revit — not Fusion 360. You will export STEP and import through the CAD Import Module, and an imported STEP body is dead geometry: no parameters, so COMSOL cannot regenerate it when a parameter changes.

If you build your simulation geometry in Fusion, your parametric sweep and your topology optimisation both become impossible. Build two geometries for two jobs.

	Fusion 360 — design model	COMSOL — simulation model
Purpose	CAD deliverable, manufacture, presentation	Parametric sweeps and optimisation
Contents	All subsystems, fasteners, gaskets, fillets, GD&T	Absorber, fluid volume, layers that carry heat
Parameters	Fusion User Parameters	COMSOL Global Parameters, rebuilt natively
Regeneration	Manual	Automatic, 300 times, unattended
Leaves its tool?	As STEP, for reference only	Never

7.1 If you must import

1. In Fusion, create a SIM configuration. Suppress fasteners, gaskets, coatings, logos, and every fillet below 1 mm EXCEPT the channel root fillet. Revision H sets r_fillet to 0.5 mm under Decision D4, so a blanket sub-1 mm suppression would remove the very feature 6.3 identifies as carrying the lateral conduction path, and would quietly change the answer. In this model the suppression set is 57 components — 30 fasteners, 5 seals, 14 mounts and 8 sensors — plus the selective coating, 58 bodies in all. Do not suppress the 12 ports; they carry the inlet and outlet faces.
2. Export STEP AP214. Never STL — faceted and useless for CFD.
3. Import with Detect interfaces on, repair tolerance 1e-5.
4. Finalize as Form Assembly on this geometry. Form Union is the general advice, because it shares solid-fluid interfaces directly, but on this model in COMSOL 6.2 it produced no usable exterior fluid boundary: every boundary reported as not applicable even with the flow interface on all domains. Form Assembly gives 3 domains, 673 boundaries and 2 identity pairs, and COMSOL adds a Continuity node on those pairs automatically. That node is what carries conjugate heat transfer across the interface, so confirm it exists rather than assuming it, because without it Form Assembly does silently insulate the fluid from the plate.
5. Run Defeaturing and Repair: Remove Small Faces, Remove Details, Delete Fillets, Remove Short Edges. Sliver faces are the top cause of mesh failure on imported CAD.
6. Rebuild every boundary selection as a Box, Ball or Cylinder selection, never by face number. Face numbering changes on re-import and will move your inlet to an arbitrary wall without warning.

After any import, run a trivial isothermal flow study: mass out must equal mass in to 0.1 %, and the fluid domain must be connected end to end. A pinched channel from a failed Boolean is invisible in the graphics window and fatal downstream.

8. CFD methodology in COMSOL 6.2

8.1 Modelling unit

Papers 2, 3 and 15 solve whole-system transient thermosiphons at 2–7 million elements. You cannot run 250 of those, and you need about 250 for a surrogate. This project uses a single-channel absorber strip at steady state with prescribed inlet mass flow.

Approach	Elements	Per run	250 runs	Verdict
Full system, transient	2–7 M	1–4 days	2–3 years	Impossible
Full system, steady state	2–5 M	3–8 h	~4 months	Too slow
Single-channel strip, steady	0.3–0.9 M	6–25 min	40–90 h	Adopt

Cite the precedent rather than hiding it: Sakib et al. used this to generate 1000 runs, citing Gunjo et al. (full ten-tube collector from one tube), Duffie and Beckman (strip analysis), and Mintsa Do Ango (single channel with symmetry planes). State both limitations explicitly — no manifold maldistribution, no edge losses. A named limitation is stronger than an unmentioned one.

Note the exception: the alternating counter-current arrangement (B1–B3) is inherently a multi-channel effect and cannot be captured on a single channel. Model a three-channel strip with symmetry for that study, roughly 2.5× the cost.

BUILT IN REVISION I. The strip above was a specification until now; it exists in the CAD model as of version 16 and this paragraph records what was actually cut, because the document and the model must not drift apart again. CFD_STRIP_3CH_MASTER spans x from -550 to +550 mm, the full channel length, and y from -80 to +40 mm, a width of 120 mm equal to three channel pitches. Both y cut planes land on bridge midlines, 11.098 mm clear of the nearest channel feature, and the solid is trimmed to the channel ends so the fluid inlet and outlet faces are coplanar with the solid end faces. Manifolds are excluded, which is the limitation this section already names. It carries channels CH05, CH06 and CH07 — odd, even, odd — so the middle channel has a counter-flowing neighbour on both sides. Solid 281,693.638 mm³, three fluid bodies of 18,643.963 mm³ each.

The geometry confirms the alternation independently of any parameter: the large channel end face of 28.055 mm² lies at x = -550 for CH05 and CH07 and at x = +550 for CH06. The conjugate interface was verified by sampling 169 points on each fluid body wall and querying containment against the solid — 169 on the boundary, none inside, none outside, on all three channels. There is no gap and no interpenetration.

A CORRECTION TO THIS SECTION. The instruction above to model the strip WITH SYMMETRY is not exact under the very arrangement it is meant to study. A symmetry plane imposes zero lateral heat flux, and it is valid only where the field is mirror-symmetric about the plane. At a bridge midline under alternating flow, the channel across the plane runs the opposite direction, so the field is glide-symmetric — mirrored in y AND reversed in x — not mirror-symmetric. Imposing an adiabatic plane there suppresses exactly the lateral conduction that Filing 1 exists to exploit, and it does so at the domain boundary.

Two ways to live with it, and the model now contains the geometry for both. Keep the three-channel strip as written and read results from the MIDDLE channel only, treating the two outer channels as buffers that absorb the boundary error — this is why three channels are cut rather than two, since three is the smallest domain containing a channel that never touches an artificial boundary. Or use CFD_STRIP_2CH_PERIODIC, which spans y from -80 to 0 mm, carries CH05 and CH06, and replaces symmetry with translational periodicity of period 80 mm. That domain is exact rather than approximate, and cheaper: solid 187,817.190 mm³ against 281,693.638 mm³, about two thirds the cell count. Its two periodic faces were measured at exactly 2200.000000 mm² each, identical rectangles in the bridge material, so the periodic pair maps without interpolation error.

The periodic domain is the better physics and the lower cost, and on the evidence above it should probably be the production domain, with the three-channel strip kept as the cross-check. That recommendation is not a decision. Settle it before the production sweep, record the choice, and do not mix results from the two domains in one surrogate training set.

8.2 Flow regime

$$Re = 4 \cdot \dot{m} / (\pi \cdot D \cdot \mu)$$

$\dot{m}$ 0.001-0.0045 kg/s total, over $n_{ch} = 12$
 D 0.0028-0.0052 m, μ 8.55e-4 Pa.s at 300 K

Re per channel = 30 to 210 -> deeply laminar

Far below the 2300 transition. Use Laminar Flow, not k-epsilon. Include this as a numbered equation; its absence is a common cause of revision.

8.3 Physics interfaces

Physics	Interface	Settings	Tier
Conjugate heat transfer	Heat Transfer in Solids and Fluids	Fluid node on channel, Solid on plate and layers	T1
Flow	Laminar Flow	Incompressible, stationary	T1
Coupling	Non-Isothermal Flow	Let COMSOL create it; never couple manually	T1
Cavity radiation	Surface-to-Surface Radiation	Diffuse, hemicube, absorber-to-glass cavity	T1
Sky loss	Surface-to-Ambient Radiation	$T_{sky} = 0.0552 \cdot T_{amb}^{1.5}$ (Swinbank), kelvin	T1
Wind convection	Heat Flux, convective	$h = 5.7 + 3.8 \cdot v_{wind}$	T1
Thin layers	Thin Layer feature	VIP, coating, graphite — avoids meshing thin solids	T1
PCM	Phase Change Material subnode (Heat Transfer in Solids)	Apparent Heat Capacity method, mushy zone 6 K; forces transient	T2
Graphite-loaded PCM	Heat Transfer in Porous Media	Local thermal equilibrium	T2
Thermotropic switching	Analytic $\tau(T)$ in the flux expression	Smoothed step, 72–78 °C, τ 0.88 → 0.40	T2
Optics	Ray Optics Module	Honeycomb angular transmittance $\tau(\theta)$	T3

8.4 Boundary conditions

Boundary	Condition	Value
Absorber top	Inward heat flux, then radiation losses	$q = G \cdot T \cdot \tau_{glz_sys} \cdot \tau_{tim} \cdot \alpha_{abs} \cdot (1 - f_{shade})$
Channel inlet	Mass flow, temperature	$\dot{m}_{ch}$, T_{in}
Channel outlet	Pressure, no viscous stress	$p = 0$ gauge

Boundary	Condition	Value
Channel walls	No slip, thermal continuity	R_bond as Thin Layer if non-zero
Strip sides	Symmetry (flow), insulation (thermal)	Valid for a periodic array
Glazing outer	Surface-to-ambient radiation + convection	eps_glz 0.88, h_wind
Rear of VIP	Convective flux	h = 3 W/m ² K, still air

8.5 Mesh

1. Fluid: Swept mesh along the channel, source face = inlet, 180 elements, element ratio 3, symmetric distribution.
2. Boundary Layers node inside the channel: 5 layers, stretch 1.2, thickness factor 2.5. Plain tetrahedra at the wall quietly cost several percent on outlet temperature.
3. Solids: Free Tetrahedral, Normal to Fine, with an Extra Fine Size node on the channel-root fillet.
4. Thin layers are not meshed at all — the Thin Layer feature handles them.
5. Check Statistics: minimum element quality > 0.10, average > 0.65.

Grid independence: four meshes roughly doubling element count, converge below 0.5 %, take the coarsest that meets it. This table appears in every reference paper and its absence is an immediate objection.

8.6 Solver

- Stationary study, Segregated solver, two groups: flow (u, p) and temperature (T).
- PARDISO direct solver at this size. Switch to GMRES with geometric multigrid only if memory becomes a problem.
- Relative tolerance 1e-4. Do not loosen it to force convergence — if it will not converge, the geometry or mesh is wrong.
- If the coupled solve stalls at high irradiance, add an Auxiliary Sweep ramping G_T from 200 to 1000 W/m². Far more robust than tolerance tuning.
- Define all outputs as Derived Values before any sweep, so only scalars are stored.

8.7 Validation

Source	Why it suits this model	Target
Sun et al., via Sakib 2026 ref 49	Steady state, single-tube compatible, eta vs mdot at 1000 W/m ² . Sakib achieved 1.58 % mean	< 3 %
Gunjo et al., Renew. Energy 106 (2017) 24	CFD plus experiment, steady state, full geometry and BCs published. The standard case	< 5 %
Freegah et al. 2025 (P2)	Transient — secondary trend check only, never quantitative	trend

Hard gate

If validation exceeds 8 %, stop before building any variants. Debug in order: pure conduction, then adiabatic flow-only, then add radiation. Isolate rather than tuning until the answer looks right. Every number in the remaining six months inherits this model's credibility.

8.8 Campaign

1. Generate the matrix in Python with `scipy.stats.qmc.LatinHypercube`; export CSV.
2. COMSOL: Study > Parametric Sweep, load from file.
3. Disable Store solution in memory. Storing 300 full 3D solutions exhausts the disk near the end of the run — the single most common way this phase is lost.
4. Target 300 attempts for 250 usable rows. Expect 10–15 % failures; keep a re-run queue and record why each failed, because the failure pattern maps a real geometric constraint.
5. Log per run: `T_out`, `dP`, `Qu`, `eta`, `T_p,max`, `R4`, convergence status, wall-clock time.

9. Topology optimisation

Conventional optimisation asks what fin height is best. Topology optimisation asks where the metal should be at all. It does not select among designs; it generates one. Applied extensively to heat sinks and once to a parabolic dish cavity receiver — never, in this search, to a flat plate absorber.

9.1 COMSOL setup

1. Definitions > Topology Optimization, Density Model method, scoped to the absorber design domain.
2. Set filter radius $R_{\min}$ to at least $2.5\times$ local mesh size and no less than the roll-bond manufacturable minimum, about 3 mm. This one parameter decides manufacturable versus fractal.
3. Projection slope beta by continuation: 1, 2, 4, 8, 16, each solution seeding the next. Starting high will not converge.
4. Add Brinkman penalisation via the Porous Medium node on Laminar Flow, inverse permeability interpolated by RAMP or SIMP of the design variable.
5. Objective: weighted thermal compliance and dissipated power, weight w_{therm} .
6. Volume constraint $\phi_{\max}$ 0.15–0.40. Without it the optimiser fills the domain.
7. Solver: MMA, 100–200 iterations. Budget 4–12 h per run; you will do about ten, not three hundred.
8. Export the filtered density field as STL, re-import it as geometry with face detection, defeature, re-mesh, and run a clean verification simulation. Node names for this step differ between COMSOL versions — confirm against your 6.2 installation (verification item V4).

The step almost everyone skips

A TO result is a density field, not a part. Extract it as geometry, smooth it, impose the manufacturing minimum feature, re-mesh, and re-run full conjugate heat transfer on the interpreted shape. Interpreted performance is always somewhat worse than the optimiser's own figure. Report both numbers and the gap between them; quoting only the optimiser's figure is caught immediately.

9.2 The stagnation constraint

Run the optimisation twice: unconstrained, and constrained so stagnation temperature must not exceed 130 °C. Compute T_{stag} from a zero-flow case — inlet mass flow effectively zero, full irradiance, solve for steady plate temperature — and feed it as a constraint.

The comparison answers a question the field has never asked: how much thermal performance must you surrender to build a collector that survives its own failure mode? Memorable, defensible, publishable, and one extra study step.

9.3 Surrogate and multi-objective search

The surrogate models themselves are covered in full in Section 10. This subsection states only how they connect to the optimiser.

Topology optimisation is too expensive to run inside a genetic algorithm. Use it to generate a few high-quality geometries; use the parametric campaign of Section 8.8 to train the surrogate models of Section 10; then run the multi-objective search against the surrogate rather than against CFD.

- NSGA-II via pymoo: population 100, 100 generations, SBX $pc = 0.9$, polynomial mutation $pm = 1/n_{\text{var}}$, $\eta_m = 20$, duplicate elimination on.
- Hold irradiance and inlet temperature fixed during optimisation.

- Close the loop: run the knee-point design back through COMSOL and report predicted versus actual. One afternoon; it separates a credible optimisation paper from a plausible one.

10. ANN and hybrid ANN modelling

This section replaces the outline in Section 9.3 with the full training programme. The models built here are consumed twice: by the NSGA-II search in Section 9, and by the Simulink collector block in Section 11.

10.1 Why a surrogate is not optional

The arithmetic decides this, not preference. One CFD run takes 6 to 25 minutes. NSGA-II with a population of 100 over 100 generations needs 10,000 evaluations.

Route	Per evaluation	10,000 evaluations	Verdict
Direct CFD in the loop	6–25 min	42–170 days	Impossible
Trained ANN surrogate	< 1 ms	under 10 seconds	Adopt

The surrogate is trained once on 250 to 350 CFD runs — a one-time investment of 40 to 90 machine-hours — and then answers in milliseconds for the rest of the project. This is the same argument Sakib et al. make, and it is worth stating explicitly in your methodology chapter because it justifies the entire machine-learning component.

10.2 Dataset preparation

Step	Action	Detail
1	Assemble	One CSV: 250–350 rows, 16 input columns (Section 4.8 active set), 6 output columns. Revision H added <code>r_fillet</code> to the active set under Decision D4, so the width is 16, not 15.
2	Outputs	<code>T_out</code> , <code>dP</code> , <code>Qu</code> , <code>eta</code> , <code>T_p,max</code> , <code>R4</code> . Train a separate model per output — multi-output networks are consistently less accurate here
3	Audit failures	Keep a separate log of failed runs with the reason. Never silently drop them; the failure pattern maps a real geometric constraint
4	Encode categoricals	<code>f_topo</code> and <code>f_interdig</code> are categorical. One-hot encode <code>f_topo</code> ; <code>f_interdig</code> is already binary
5	Split	70 / 15 / 15 train / validation / test, stratified on <code>f_topo</code> so every flow-field type appears in all three splits
6	Scale	StandardScaler fitted on the TRAINING SPLIT ONLY, then applied to validation and test. Fitting on the full set leaks information and reviewers check for it
7	Store	Save the fitted scaler alongside the model. Without it the model cannot be used later, in Simulink or anywhere else

The single most common error in this kind of study

Fitting the scaler on all the data before splitting. It inflates test-set performance because the test rows influenced the mean and standard deviation the model was trained against. It is invisible in your results, it makes your R^2 look better than it is, and it is the first thing an examiner familiar with machine learning will check. Fit on train, transform everything.

10.3 Baseline ANN

Build this first and get it working before attempting any hybrid. It is also your control: every hybrid must beat it, and if none does, that is a legitimate and reportable result.

Setting	Value	Reason
Architecture	16 → 64 → 32 → 16 → 1	Tapering depth suits a smooth, low-noise response surface from CFD
Hidden activation	ReLU	Standard; try tanh as an alternative for the smoother outputs
Output activation	Linear	Regression, not classification
Weight initialisation	He normal	Matched to ReLU
Optimiser	Adam, initial learning rate 1e-3	Robust default
Learning rate schedule	ReduceLROnPlateau, factor 0.5, patience 20	Lets the model settle without manual tuning
Batch size	16	Small dataset; large batches under-fit here
Loss	Mean squared error	Penalises the large errors that matter for optimisation
Regularisation	L2 1e-4 on all dense layers	Essential at 250 rows
Early stopping	Patience 50, restore best weights, max 1000 epochs	Prevents overfitting without guessing epoch count
Repeats	Train 10 times with different seeds	Report mean and standard deviation, never a single lucky run

Hyperparameter search

Run a small randomised search on the validation split, not the test split. Fifty trials is enough at this dataset size.

Hyperparameter	Search range
Hidden layers	2 to 4
Neurons per layer	16, 32, 64, 128
Learning rate	1e-4 to 1e-2, log-uniform
L2 coefficient	1e-6 to 1e-2, log-uniform
Batch size	8, 16, 32
Activation	ReLU, tanh, ELU

10.4 Hybrid ANN models

Hybrid means the network is combined with a second technique. Four variants are worth building; the first two are conventional in this literature and easy to defend, the third is the interesting one, the fourth is nearly free.

ID	Hybrid	What is hybridised	Effort	Priority
H1	GA-ANN	A genetic algorithm searches the weights and biases instead of, or before, backpropagation	Medium	Build
H2	PSO-ANN	Particle swarm optimisation trains the weights; usually faster to converge than GA	Medium	Build

ID	Hybrid	What is hybridised	Effort	Priority
H3	PINN — physics-informed	Physical residuals added to the loss so predictions obey conservation	High	Build — the novel one
H4	Stacked ensemble	ANN, XGBoost and GPR predictions combined by a ridge meta-learner	Low	Build
H5	ANFIS	Neuro-fuzzy inference; interpretable rules but scales poorly past ~8 inputs	Medium	Optional

H1 and H2 — metaheuristic weight training

Backpropagation is a local search and can settle into a poor minimum on a small dataset. A genetic algorithm or particle swarm searches globally instead. Two usable configurations:

- Full replacement: the metaheuristic optimises every weight and bias directly. Simple to explain, slow to converge, and struggles beyond a few thousand parameters.
- Hybrid two-stage, which is what to actually do: the metaheuristic finds good initial weights over 50 to 100 generations, then Adam refines them. Faster and consistently better than either alone.

Settings that work: GA with population 50, tournament selection, crossover 0.8, mutation 0.05, 100 generations. PSO with 40 particles, inertia 0.7, cognitive and social coefficients both 1.5, 150 iterations. Fitness in both cases is validation-set RMSE, not training RMSE.

H3 — physics-informed neural network

This is the hybrid worth building for its own sake, and it is a genuine methodological contribution because no paper in your reference set does it. A plain ANN can predict an outlet temperature and a useful heat gain that are mutually inconsistent — it has no knowledge that energy is conserved. A physics-informed network is penalised when it does.

```

L_total = L_data + lambda_1 * L_energy + lambda_2 * L_curve

L_data = MSE( predicted , CFD )

L_energy = MSE( Qu_pred - mdot*cp*(T_out_pred - T_in) )
           enforces the first law on every prediction

L_curve = MSE( eta_pred - [eta_0 - a1*x - a2*x^2] ), x = (T_m-T_a)/G
           enforces the Hottel-Whillier-Bliss form

lambda_1, lambda_2 : start at 0.1, tune on the validation split

```

Three benefits worth claiming. Predictions conserve energy by construction, which matters because the NSGA-II objectives are computed from them. Accuracy improves in sparse regions of the design space, because physics supplies information where data does not. And the fitted efficiency coefficients feed straight into Section 11.2, so the physics-informed model and the Simulink model become mutually consistent rather than two independent fits.

H4 — stacked ensemble

Train ANN, XGBoost and Gaussian process regression on the same split, then fit a ridge regression on their validation predictions to produce the final output. Costs almost nothing once the individual models exist and reliably gains one to three percent on RMSE. Include it because it is cheap, but do not build the thesis around it.

10.5 Benchmarks to include

Your five reference papers set the bar. Paper 13 used a Kolmogorov–Arnold network and reported R^2 of 0.992 and 0.993, so a study that reports only a plain MLP will look dated beside it.

Model	Role	Note
Multiple linear regression	Floor	Establishes that the problem is nonlinear at all
2nd-order polynomial	Classical baseline	Sakib's Table 8 shows it is competitive; include for honesty
k-nearest neighbours	Simple baseline	Cannot extrapolate; piecewise-constant surface. Benchmark only, never the optimisation surrogate
Random forest / XGBoost	Strong baseline	Also supplies feature importance almost free
Gaussian process regression	Strong baseline	Gives calibrated uncertainty, which none of the others do
Baseline ANN	Control	Every hybrid must beat this
GA-ANN, PSO-ANN	Hybrid	The conventional hybrids in this literature
PINN	Hybrid	The novel one
KAN	Reference-set benchmark	Optional but strongly advised — it is what paper 13 used, and beating it is a headline result

10.6 Evaluation and acceptance

Metric	Definition	Target
R^2	Coefficient of determination on the held-out test split	≥ 0.97 , ideally ≥ 0.99
RMSE	Root mean square error, in the physical unit of the output	T_out below 0.5 K; dP below 0.15 Pa
MAE	Mean absolute error	Report alongside RMSE; a large gap indicates outliers
MAPE	Mean absolute percentage error	Report, but treat with care where an output approaches zero
Energy residual	Mean of $ Q_u - \dot{m} \cdot c_p \cdot \Delta T $ across the test set	PINN should be an order of magnitude below the plain ANN
Seed spread	Standard deviation of R^2 across 10 seeds	Below 0.005; a larger spread means the model is unstable

Report every metric on the held-out test split only. Training and validation metrics belong in the appendix, not the results table.

Gate

If no model reaches R^2 of 0.97 on the test split, do not proceed to NSGA-II. The cause is almost always too few usable CFD rows or a design space too wide for the sample count. Fix it by adding runs in the region the surrogate predicts worst — an adaptive sampling loop of 30 to 50 extra runs typically closes the gap — rather than by accepting a weak surrogate and optimising against its errors.

10.7 Interpretability

Accuracy alone is not a result. Show that the model learned physics rather than noise.

1. Compute permutation importance and SHAP values on the test split for the best model.

2. Compute Pearson correlation and significance between every input and every output on the raw dataset, independently of any model.
3. Compare the two rankings. Where they agree, say so explicitly — that agreement is real evidence the surrogate is physically sensible, and it is the single strongest move in Sakib's paper.
4. Where they disagree, investigate. A model importance with no correlation counterpart usually indicates an interaction effect, which is worth a paragraph of its own.

Expected outcome, based on the physics: $\dot{m}$, Dh_{in} and G_T should dominate outlet temperature; Dh_{in} , $\dot{m}$ and ap_L should dominate pressure drop; w_{bridge} and $f_{interdig}$ should dominate R4 and appear nowhere else. That last prediction is a falsifiable test of the counter-current mechanism — if w_{bridge} shows up as important for R4 and unimportant for everything else, the mechanism is behaving as theorised.

10.8 Deployment

- Export the trained model and its scaler together. Pickle or joblib for Python; ONNX if you want portability.
- For Simulink (Section 11), either export weights to a MATLAB Function block, or retrain the final model in MATLAB using the Deep Learning Toolbox so it drops straight into the model. Retraining in MATLAB is usually less painful than bridging Python and Simulink.
- Report inference time per evaluation; it is the number that justifies the whole approach.
- Version the dataset alongside the model. A model without the exact CSV that trained it is not reproducible, and reproducibility is a required section in most target journals.

10.9 Failure modes and fixes

Symptom	Likely cause	Fix
Test R^2 far below validation R^2	Scaler fitted before the split, or the test split is unrepresentative	Refit the scaler on train only; stratify the split on f_{topo}
Training loss above validation loss	Normal — regularisation applies during training only, and training loss is a running mini-batch average	No action; explain it in the text as Sakib does
Loss oscillates and will not settle	Learning rate too high, or batch size too small	Halve the learning rate; enable <code>ReduceLROnPlateau</code>
Excellent metrics, poor optimisation results	The surrogate is accurate on average but wrong at the design-space edges where the optimiser goes	Adaptive sampling: add CFD runs at the optimiser's proposed optima and retrain
Hybrids no better than the baseline	A legitimate result on a smooth, low-noise CFD dataset	Report it honestly. Negative results about hybrid methods are publishable and rarer than positive ones
Wildly different results between seeds	Dataset too small for the architecture	Reduce network size before adding data; 15 inputs do not need 128 neurons per layer

11. MATLAB and Simulink system model

CFD gives component performance at a point. Simulink gives system performance over a year. The two are complementary and the link between them is a three-coefficient efficiency curve.

11.1 Why this section exists

Gap 11 in Section 2.1: no reference paper reports annual yield, part-load behaviour or off-design performance. All five report a single day. The Simulink model closes that gap, and it is also where you prove the process-heat argument of Section 1 and generate the stagnation statistics that support the patent claim.

Question	Answered by
What is the efficiency at this operating point?	COMSOL CFD
What shape should the flow field be?	COMSOL topology optimisation
How much hot water does this deliver in a year in Chennai?	Simulink
How many hours per year does it exceed 130 °C?	Simulink
Does it still work on an 80 °C process load?	Simulink
What collector area gives 70 % solar fraction?	Simulink

11.2 The CFD-to-Simulink bridge

Fit the standard quadratic efficiency curve to your CFD results and pass only the three coefficients into Simulink. This is the same form used by EN 12975 and by every commercial simulation tool, so your collector becomes directly comparable to certified products.

$$\eta = \eta_0 - a_1(T_m - T_a)/G - a_2(T_m - T_a)^2/G$$

$$T_m = (T_{in} + T_{out})/2 \quad \text{mean fluid temperature}$$

Fit from ≥ 12 CFD points spanning:

$$(T_m - T_a)/G = 0 \text{ to } 0.08 \text{ m}^2\text{K/W} \quad \text{at } G = 800 \text{ W/m}^2$$

Expected: conventional η_0 0.78, a_1 6.75, $a_2 \sim 0.010$. GRAIL η_0 0.66, a_1 2.00, $a_2 \sim 0.004$ for the single-glazed stack; with the double glazing as built the optical term alone takes η_0 to about 0.60, while a_1 falls by an amount not yet computed. Fit both terms from your own CFD rather than carrying either figure forward – see correction C11 and Section 1.

$$\text{GRAIL} \quad \eta_0 \text{ 0.66, } a_1 \text{ 2.00, } a_2 \sim 0.004$$

Report the fit quality ($R^2 > 0.99$ is achievable) and the residuals. The fitted curve, not the raw CFD, is what the annual model consumes.

11.3 Model architecture

#	Block	Implementation	Key settings
1	Weather input	From Spreadsheet block reading a TMY file	ASHRAE or EnergyPlus .epw for your city, hourly, 8760 rows
2	Solar geometry	MATLAB Function block	Declination, hour angle, incidence on a 33° tilted plane

#	Block	Implementation	Key settings
3	Incidence angle modifier	Lookup table	$IAM = 1 - b_0 \cdot (1/\cos(\theta) - 1)$, $b_0 \approx 0.1$ for TIM glazing
4	Collector	MATLAB Function or Simscape thermal-liquid	Quadratic curve from Section 11.2 plus thermal capacitance C_{col} for dynamics
5	Piping	Simscape Fluids pipe	Length, insulation, ambient loss
6	Stratified tank	Simscape or S-Function, N-node	5–10 nodes, 200 L, includes internal heat exchanger
7	Draw-off load	Signal Builder or From Spreadsheet	EN 12976 profile, or 200 L/day at 45 °C for DHW
8	Controller	Stateflow chart	Differential thermostat: ON at ΔT 8 K, OFF at ΔT 3 K, plus high-limit cutout
9	Auxiliary heater	Saturation + gain	Tops up to setpoint; its energy is what the solar fraction displaces
10	Stagnation detector	Stateflow	Latches when pump off AND $T_{plate} > T_{stag_max}$; counts events and hours

Toolboxes: Simulink is essential. Simscape and Simscape Fluids make the hydraulic loop and tank far easier than hand-coded S-Functions and are worth using if licensed. Stateflow for the controller. Deep Learning Toolbox only if you train the ANN in MATLAB rather than Python — either is fine, but pick one and stay there.

11.4 Two operating cases

Case	Load	Setpoint	What it demonstrates
A — domestic hot water	200 L/day, EN 12976 profile	45 °C	Annual yield and solar fraction against a conventional collector
B — process heat	Continuous, constant return temperature	80 °C	The regime argument: conventional collapses, GRAIL does not. This is the strongest result in the thesis

Run both for a full 8760-hour year, for both collectors, in the same weather file. Four annual simulations, each a few minutes. This is the cheapest high-value result in the entire project.

11.5 Validation of the Simulink model

1. Static check: at fixed inputs, the Simulink collector block must reproduce the COMSOL efficiency at the same operating point to within 2 %. If it does not, the curve fit or the unit conversions are wrong.
2. Energy balance: over one simulated year, collected energy minus losses minus delivered energy must close to within 1 %.
3. Sanity check against published yields: a good 0.55 m² flat plate in an Indian climate delivers roughly 350–550 kWh/m²/year for DHW duty. If your model reports 1200, something is wrong.
4. Timestep independence: run at 60 s, 300 s and 900 s. Annual yield should change by under 1 %. Report it, exactly as you report grid independence for the CFD.

12. Results and figures to produce

A complete checklist. Priority CORE means it must be in the thesis and the paper; SUPP means include if space allows or put in an appendix.

12.1 Figures

#	Figure	Content	Priority
F1	Grid independence	T _{out} and dP vs element count, 4 meshes	CORE
F2	Validation curve	eta vs (T _i -T _a)/G, present CFD vs Gunjo/Sun, with error bars	CORE
F3	Efficiency curves	Conventional vs GRAIL, both fitted quadratics, crossover marked	CORE
F4	Geometry family	Rendered: straight, graded, counter-current, TO	CORE
F5	Temperature contours	Absorber surface, four designs, identical colour scale	CORE
F6	Velocity field	Streamlines and contours in the channels	SUPP
F7	Plate temperature profile	T along flow direction, all designs on one axis	CORE
F8	Counter-current, side by side	Plate temperature field, parallel vs counter-current, same scale	CORE — key figure
F9	Temperature histogram	Distribution of plate temperature, both arrangements — shows the flattening	CORE
F10	R4 vs bridge width	Radiative proxy against w _{bridge} , showing the interior optimum	CORE
F11	Loss breakdown	Stacked bars: radiative, convective, rear — both arrangements	CORE
F12	Correlation heatmap	Pearson r, all inputs against all outputs, with significance	CORE
F13	Grading result	eta vs G _{ratio} at several lam _G	CORE
F14	Grading penalty	dP vs G _{ratio} , same series	CORE
F15	Response surfaces	2D slices for the three most influential parameters	SUPP
F16	Parity plots	ANN, KNN, RF, polynomial — 2 × 4 grid, both outputs	CORE
F17	Training curves	Loss vs epoch, train and validation	SUPP
F18	Model comparison	Bar chart: R ² , RMSE, MAE, MAPE across four models	CORE
F19	TO convergence	Objective vs iteration, beta continuation steps marked	CORE
F20	TO evolution	Density field at iterations 1, 20, 60, 120	CORE
F21	Optimised geometry	Raw density field, interpreted CAD, and re-meshed model	CORE
F22	TO performance	Optimised vs graded vs straight, with the optimiser-versus-interpreted gap shown	CORE

#	Figure	Content	Priority
F23	Stagnation fields	T_stag contours, unconstrained vs constrained optimum	CORE
F24	Stagnation trade-off	eta against T_stag_max — the original contribution	CORE — key figure
F25	Stagnation transient	Plate T vs time after flow stoppage, both collectors	SUPP
F26	Pareto front	Qu vs dP, knee marked, generations 1/10/50/100 overlaid	CORE
F27	Pareto variables	Distribution of each design variable across the front	CORE
F28	Annual yield	Monthly bars, conventional vs GRAIL, DHW case	CORE
F29	Solar fraction	Against collector area, both collectors	SUPP
F30	Process heat case	Annual yield at 80 °C load — the regime argument	CORE — key figure
F31	Stagnation hours	Hours per year above 130 °C, both collectors	CORE
F32	Carbon payback	Cumulative avoided CO ₂ against embodied carbon, crossing point marked	SUPP
F33	Hybrid model comparison	Baseline ANN vs GA-ANN, PSO-ANN, PINN, stacked ensemble — R ² and RMSE	CORE
F34	Energy residual	Qu - mdot·cp·ΔT distribution, plain ANN vs PINN — proves conservation	CORE
F35	SHAP importance	SHAP values per input, compared against Pearson correlation ranking	CORE

Twenty-nine are marked CORE. That is more than a journal paper will hold — expect to publish about twelve and keep the rest in the thesis. The four labelled key figures are F8, F24, F30 and F3; if a reader remembers only four images, those are the four that carry the contribution.

12.2 Tables

#	Table	Content
T1	Grid independence	4 meshes: elements, T_out, dP, % change, wall time
T2	Validation	Point-by-point CFD vs published, % deviation, mean and max
T3	Efficiency coefficients	eta_0, a1, a2 with fit R ² , both collectors — feeds Simulink
T4	Design parameter set	The active sweep set with ranges, as Section 4
T5	Surrogate performance	R ² , RMSE, MAE, MAPE for the baseline ANN and every benchmark, both outputs, held-out test set
T6	TO settings	R_min, beta schedule, phi_max, iterations, wall time
T7	Optimum comparison	Straight vs graded vs counter-current vs TO: eta, dP, T_stag, R4, mass
T8	Knee point verification	Surrogate prediction vs CFD actual, % error
T9	Annual performance	Yield, solar fraction, stagnation hours, both cases, both collectors

#	Table	Content
T10	Timestep independence	Simulink annual yield at 60, 300, 900 s
T11	Comparison with literature	Your gain vs the five reference papers, on a normalised basis with baselines stated
T12	Hybrid ANN comparison	Baseline ANN against GA-ANN, PSO-ANN, PINN and the stacked ensemble: R^2 , RMSE, MAE, MAPE, energy residual, training time, seed spread

12.3 The three headline numbers

Every thesis needs a small number of results a reader can carry away. Aim to be able to complete these three sentences:

1. The alternating counter-current arrangement reduced the radiative loss proxy R_4 by ____ % at unchanged mean plate temperature and unchanged mass flow, raising thermal efficiency by ____ points.
2. The topology-optimised absorber delivered ____ % more useful heat than the straight-channel baseline at equal pressure drop, and ____ % more than the axially graded design.
3. Under an 80 °C process-heat duty in [city], the GRAIL collector delivered ____ kWh/m²/year against ____ for a conventional collector, while spending ____ fewer hours above 130 °C.

13. Patent strategy

An informed engineering assessment for project planning, not legal advice. A registered patent agent must review any filing.

13.1 Two different things in this document are called patents

This distinction matters and it is easy to conflate the two. Both appear in this report and they mean opposite things.

	Our filings	Third-party patents
What they are	Inventions arising from THIS project, drafted as claim language for you to file	Patents owned by other organisations, found in the prior-art search
Where	Sections 13.2 and 13.6 — three filings	Section 13.4 — the landscape table
Why they are here	These are your intellectual property, if you file before publishing	These are obstacles and context: what you cannot claim, and what you must design around
Example	Filing 1, the alternating counter-current absorber, comes directly from parameters B1–B3 of your own design	FracTherm is Fraunhofer's. You cannot claim it and would need a licence to commercialise it
Count	3 filings proposed	22 entries, 26 patent documents

To answer the question directly: yes, every claim drafted in Section 13.6 derives from this project's own design parameters — Filing 1 from Group B, Filing 2 from parameters C3 and C4, Filing 3 from the PCM subsystem. Nothing in Section 13.6 is copied or adapted from a third-party patent. The third-party patents in Section 13.4 exist in this document solely to establish what is already taken.

13.2 The three filings from this project

#	Filing	From parameters	Strength	When
1	Alternating counter-current absorber	B1, B2, B3	STRONGEST	Week 20
2	Spatially patterned static emittance zones	C3, C4	Moderate, narrow	Week 20
3	Axially graded thermal capacitance	PCM subsystem	Moderate	Week 20
+	Topology-optimised geometry, as an article	B4, B5, F1–F4	Depends on result	Complete spec, wk 24+

Filing 1 — Alternating counter-current absorber

Adjacent channels convey fluid in opposite directions, so a hot outlet-end region of one channel lies laterally adjacent to a cold inlet-end region of its neighbour, coupled by conduction through the plate.

The technical effect is provable. Radiative loss scales as the fourth power of absolute surface temperature, so by Jensen's inequality a flatter field at fixed mean has a strictly lower mean of the fourth power of temperature. The arrangement reduces radiative loss at unchanged mean temperature, unchanged mass flow and unchanged heat transfer coefficient. It improves the collector without improving heat transfer at all.

The prior art uniformly teaches the opposite approach — improve heat transfer into the fluid — and contains no suggestion of temperature-field homogenisation as a loss-reduction strategy. It passes Section 3(f) because the channels do not function independently, and Section 3(k) because it is an article, not a method. Evidence: figures F8, F9, F10, F11.

Filing 2 — Spatially patterned static emittance zones

A single absorber plate carrying two fixed emittance values in different zones: low emittance over most of the area, and a higher-emittance relief zone adjacent the outlet where the plate runs hottest. In normal operation the relief zone costs little because it is small and the plate is cool. At stagnation, when the whole plate goes hot, it becomes a passive radiative relief valve.

Search result: no prior art located for spatial patterning. All located work is either spectral optimisation across wavelength (US7607428B2, US20100313875A1, FR3023362A1) or thermochromic switching across time (EPFL, VO2:Ge). The claim must therefore recite spatial patterning and expressly disclaim thermochromic and switchable materials — that disclaimer is what keeps it clear of the established art.

Filing 3 — Axially graded thermal capacitance

The latent heat storage layer behind the absorber is not uniform. Areal PCM mass increases monotonically toward the outlet, concentrating buffering capacity where plate temperature is highest and where stagnation initiates.

Search result: every located PCM-in-collector study uses a uniform layer or uniform encapsulation. Non-uniform axial distribution matched to the plate temperature profile was not located. The effect is a lower peak plate temperature and a lower stagnation temperature for less total PCM mass than a uniform layer — a genuine efficiency in material use, and it stacks with Filing 1 because both flatten the same temperature field.

13.3 Critical terminology warning

Do not use the word interdigitated in your claims

A search of the term found that interdigitated flow field is established, heavily patented terminology in PEM fuel cells and redox flow batteries — US9166243B2 (flow battery with interdigitated flow field), US7838139B2 (flow field plate geometries), US20150030964 (fuel cell component with interdigitated flow field configuration), among many others.

In those fields the word means something different from what you intend: dead-ended inlet and outlet channels that force reactant through a porous diffusion layer. It is a mass-transport device. Your arrangement has continuously open channels flowing in opposite directions, and its purpose is in-plane thermal conduction. The mechanisms are unrelated.

But an examiner searching the word will surface that art immediately, and you will spend the prosecution arguing about it. Rename the concept to alternating counter-current channel arrangement, or bidirectional counter-flow absorber, throughout the thesis and the claims. Then add one paragraph to the specification expressly distinguishing fuel-cell interdigitated flow fields, so the examiner sees you considered and excluded them.

This rename has been applied throughout Revision E. Parameter B1 keeps the symbol f_{interdig} for continuity with earlier data files, but should be described in writing as the alternating counter-current flag.

13.4 Third-party patent landscape

Twenty-two entries covering twenty-six patent documents, catalogued across the design. Barrier means you cannot claim or practise it freely. Context means it is prior art you must cite and distinguish. Terminology means it is not technically relevant but will be surfaced by a keyword search.

Patent / family	Title and owner	Relevance	Class
FracTherm (Fraunhofer ISE)	Bionic branching channel algorithm, roll-bond absorbers; licensed to CGA Technologies 2013	Kills the branching manifold entirely	BARRIER
EP2313705A1	Roll-bond absorber	Roll-bond absorber construction for solar	BARRIER
US9513032B2	Method for producing a direct flow aluminium absorber for a solar thermal collector	Roll-bond manufacturing route	BARRIER
US20100236543A1	Method for the production of an absorber plate for solar collectors	Absorber plate manufacture	CONTEXT
US20100018521A1	Solar collector with foil absorber	Thin-sheet absorber constructions	CONTEXT
DE10037088C2	Foil solar collector with transparent thermal insulation	TIM plus foil absorber combination	BARRIER
EP2522927A2	Solar thermal collector with transparent insulation	TIM in a flat plate collector	BARRIER
CA2283890C / WO2001023813A1	Honeycomb transparent insulation with improved insulating ability	Honeycomb TIM structures	BARRIER
ES2431470A2	Thermal solar collectors with transparent insulation	TIM collectors	CONTEXT
US10889501B2	Solar thermal aerogel receiver and materials therefor	Aerogel front insulation	CONTEXT
PT1592929E	Method and apparatus for solar collector with integral stagnation temperature control	Kills mechanical stagnation limiters	BARRIER
ES2650290A1	Passive protection system against overheating of thermal solar plates	Passive stagnation protection	BARRIER
WO1981000613A1	Improved vented solar panel	Venting for overheat control	CONTEXT
US8109087B2	Overheating protection for shape memory alloy actuators	SMA overheat actuation generally	CONTEXT
US7607428B2	Solar absorber	Selective absorber constructions	CONTEXT
US20100313875A1	High temperature solar selective coatings	Selective coating chemistry	CONTEXT
FR3023362A1	Selective solar absorber with passivation layer and thermal barrier	Selective coating stacks	CONTEXT
US4330344	Selective absorber of solar energy and process for producing same	Foundational selective coating art	CONTEXT
US9166243B2	Flow battery with interdigitated flow field	Not technically relevant — but owns the keyword	TERMINOLOGY

Patent / family	Title and owner	Relevance	Class
US7838139B2	Flow field plate geometries	Fuel cell flow fields — keyword collision	TERMINOLOGY
US20150030964	Fuel cell component having an interdigitated flow field configuration	Fuel cell — keyword collision	TERMINOLOGY
EP2565554A1 / DE102008037922B4 / US5692491A / CA1034455A	Thermal and PVT collectors; tandem flat plate; transpired collector; reducing absorber heat loss	General collector loss-reduction art to cite and distinguish	CONTEXT

Read the barrier rows as design constraints, not defeats. Two of them are already resolved: drop the branching manifold and FracTherm and EP2313705A1 stop mattering for your claims; drop the mechanical vent and PT1592929E and ES2650290A1 stop mattering. The TIM patents (EP2522927A2, CA2283890C, DE10037088C2) mean you cannot claim transparent insulation itself — which is why TIM appears only in the preamble of Filing 1's claim, as known context, never in the characterising portion.

13.5 The two statutory exclusions that govern everything

Section	In substance	Consequence
3(f)	A mere arrangement or re-arrangement of known devices each functioning independently is not an invention	A combination claim survives only if the elements interact. Bolting TIM onto a known absorber fails. TIM creating a hazard resolved by a co-designed flow field does not
3(k)	Mathematical or business methods, computer programmes per se, and algorithms are not inventions	The TO method and the stagnation-constrained procedure are excluded. The geometry they produce is not. Claim the article, publish the method

13.6 Draft claims

Engineering sketches to give your patent agent a starting point. Not attorney-drafted.

Filing 1 — independent claim

1. A solar thermal collector comprising:

- (a) a monolithic absorber panel formed of two metal sheets joined by metallurgical bonding and defining a plurality of internal fluid channels therebetween, said panel having a solar-facing surface bearing a spectrally selective coating;
- (b) an inlet manifold and an outlet manifold in fluid communication with said plurality of channels;

characterised in that said plurality of channels comprises a first set and a second set arranged in alternating lateral adjacency, the first set conveying working fluid in a first direction and the second set in a direction substantially opposite thereto, each channel being continuously open between its manifolds, such that each channel of the first set is laterally adjacent along substantially its whole length to at least one channel of the second set, and the sheet material between laterally adjacent channels of opposite sets forms a conductive bridge of width selected to reduce the peak-to-mean temperature difference of said solar-facing surface relative to a parallel-flow arrangement of equal channel count, cross-section and

mass flow rate.

The phrase continuously open between its manifolds is deliberate: it distinguishes the dead-ended fuel-cell interdigitated geometry of US9166243B2 and US7838139B2 in the claim itself, not merely in the description.

Dependent claims:

1. Wherein hydraulic diameter decreases monotonically along the flow direction per $Dh(x) = Dh_{in} \cdot [1 - (1 - G)(x/L)^\lambda]$, G between 0.35 and 1.00.
2. Wherein lateral pitch varies along the flow direction such that local heat flux density remains substantially uniform.
3. Wherein said conductive bridge has a width between 20 and 60 mm.
4. Wherein the collector further comprises a transparent insulation layer of through-thickness conductivity below 0.10 W/m.K between the solar-facing surface and an outer glazing, and exhibits a stagnation temperature not exceeding 130 °C at 1000 W/m² and 40 °C ambient.
5. Wherein said channels define a flow field of branching density increasing monotonically toward the outlet, minimum feature size not less than 3 mm.

Filing 2 — independent claim

1. A solar thermal absorber panel comprising a substrate having a solar-facing surface bearing a spectrally selective coating,

characterised in that said solar-facing surface comprises a first region having a fixed thermal emittance e_1 and a second region having a fixed thermal emittance e_2 , wherein e_2 is at least five times e_1 , neither coating being thermochromic or otherwise temperature-switchable, and said second region is located adjacent the working-fluid outlet and occupies between 5 and 25 per cent of the aperture area, whereby the stagnation temperature of the panel is reduced relative to a uniformly coated panel of equal solar absorptance and equal aperture area.

The express disclaimer of thermochromic and switchable coatings is not optional. Without it the claim reads onto the EPFL and VO2:Ge art and will be refused.

Filing 3 — independent claim

1. A solar thermal collector comprising an absorber panel having a working-fluid inlet and outlet, and a latent heat storage layer in thermal contact with a rear face of said panel,

characterised in that the areal mass of phase change material per unit absorber area increases monotonically in the direction of working fluid flow, from a first value adjacent the inlet to a second value adjacent the outlet that is at least 1.5 times said first value, whereby latent buffering capacity is concentrated in the region of highest absorber surface temperature and the peak absorber temperature is reduced relative to a collector having a uniform layer of equal total phase change material mass.

Note the comparator: equal total PCM mass, not equal thickness. That framing is what makes the claim an efficiency in material use rather than simply more PCM, and it is the difference between an inventive step and an obvious one.

Two drafting rules that decide whether these survive examination

The characterising portion must recite the interaction, not the parts. An examiner applying 3(f) looks for whether elements function independently; Filing 1 survives because the bridge and the opposed flow directions have no meaning separately.

Every performance number you recite must be demonstrated in the specification. If you claim a 130 °C ceiling, the description must contain the simulation that shows it. Unsupported numerical limitations are the commonest reason combination claims are narrowed on examination.

13.7 Design registration and freedom to operate

Aspect	Detail
Statute	Designs Act 2000 — protects shape and configuration judged by the eye, not function
Why it fits	A topology-optimised absorber has a distinctive visual form — exactly what design registration exists for
Term and fee	10 years renewable by 5; about ₹1,000 for an individual applicant
Timeline	Months, not years. Far faster and more certain than a patent. File it regardless of the patent outcome
No middle option	India has no utility model system, unlike China or Germany
FTO — branching manifold	FracTherm and EP2313705A1. Resolved by dropping the manifold; Filing 1 does not need branching
FTO — roll-bond process	US9513032B2 and EP2313705A1 cover roll-bond absorber production. For an academic study this is irrelevant; for commercialisation you would buy plates from a licensed producer rather than making them
FTO — transparent insulation	EP2522927A2, CA2283890C, DE10037088C2. TIM appears only in the preamble as known context, never in a characterising portion
FTO — commercial components	TIM panels, VIPs and thermotropic films are bought on the open market; purchase exhausts the relevant rights

13.8 Search strings to run before filing

Filing 1:

(solar AND collector AND absorber) AND ("counter-current" OR counterflow OR "opposite flow direction" OR bidirectional)
 (absorber AND plate) AND ("temperature uniformity" OR "temperature homogenis*" OR "peak temperature reduction")
 CPC: F24S 10/70, F24S 10/75, F24S 80/00, F24S 10/50
 Exclude: H01M (fuel cells) to suppress the keyword collision

Filing 2:

(absorber OR coating) AND ("graded emittance" OR "patterned emissivity" OR "zoned emittance" OR "regions of different emissivity") NOT thermochromic NOT switchable
 CPC: F24S 70/00, C23C 14/00

Filing 3:
 (solar AND collector) AND ("phase change") AND (graded OR
 "non-uniform" OR "varying thickness" OR "increasing toward")
 CPC: F24S 60/00, F28D 20/02

Topology-optimised geometry:
 ("topology optimi?ation" OR "generative design") AND
 (solar OR collector OR absorber)

Run on Espacenet, InPASS, Google Patents, Scopus and Web of Science. Record every string, database, date and hit count in a thesis appendix. A documented search is evidence; an undocumented claim of novelty is not.

13.9 Costs and timing

Stage	Fee ₹	Note
Provisional filing (up to 30 pp, 10 claims)	1,600	Reduced band: natural person, startup, MSME, educational institution. File Form 28
Three provisionals, if filed separately	4,800	Or combine into one filing if the agent advises unity of invention is arguable
Complete specification	1,600	Within 12 months of provisional
Request for examination	4,000	Deadline 31 months from priority
Design registration	~1,000	Separate and much faster
Agent drafting and prosecution	30,000+	Not an official fee. Your IPR cell may absorb it

Discuss with your agent whether the three filings should be one application or three. Filings 1 and 3 both act on the absorber temperature field and may satisfy unity of invention as a single inventive concept; Filing 2 is more clearly separate. One combined provisional at ₹1,600 preserves priority on all three and the decision can be deferred to the complete specification.

The one irreversible mistake

Publication destroys novelty. A conference presentation, a repository upload, a ResearchGate post or a journal submission before filing makes the invention prior art against itself, everywhere that matters. India's Section 31 grace period is narrow and does not help abroad. File the provisional first — ₹1,600 and a few days — then publish freely.

13.10 Action sequence

1. Week 4 — meet the IPR cell. Process, fee coverage, empanelled agent. Do this before you have results.
2. Weeks 4–6 — run the Section 13.8 searches formally and log them, including the H01M exclusion for Filing 1.
3. Weeks 9–18 — simulate the counter-current arrangement early, as parameter B1 in the main campaign. It is your strongest claim and you need F8–F11 to support it. Simulate graded PCM in the same campaign to support Filing 3.
4. Week 20 — draft the provisional covering Filings 1, 2 and 3. Rename interdigitated to alternating counter-current everywhere first.
5. Week 22 — file. Nothing public before this date.
6. Week 26 — file design registration on the optimised absorber form.

7. Within 12 months — complete specification, adding the topology-optimised geometry as an article claim.

14. Schedule and risk

Weeks	Phase	Output	Gate
1–3	Foundation: literature matrix, scope lock, prior-art search, IPR cell	Review, scope document, search log	Guide agrees the gap is real
4–8	Baseline COMSOL model and validation	Validated .mph, grid independence, validation plot	HARD: error < 8 %
9–12	Parametric geometry, corner testing	Parametric model, corner log	All 8 corners mesh
13–18	CFD campaign, 300 attempts, counter-current arrangement included	Dataset, 250+ rows, F5–F14	≥ 200 usable rows
19–22	ANN and hybrid ANN: baseline, GA-ANN, PSO-ANN, PINN, ensemble	F16–F18, F33–F35, T5, T12	Test $R^2 \geq 0.97$
20–22	Provisional patent drafted and filed	Filed provisional	Before any publication
22–25	Topology optimisation, constrained and unconstrained	F19–F24, T6–T7	Interpreted geometry verified
24–26	NSGA-II Pareto and knee verification	F26–F27, T8	Knee verified in COMSOL
25–28	MATLAB/Simulink annual model, both cases	F28–F31, T3, T9–T10	Energy balance closes to 1 %
27–30	Thesis, manuscript, design registration	Thesis, paper, design filing	—

14.1 Risk register

Risk	Likelihood	Mitigation
Validation exceeds 8 %	High	Two datasets ready. Debug in order: conduction, adiabatic flow, then radiation. Isolate before tuning
TO does not converge	High	Beta continuation is mandatory. Start pseudo-3D before full 3D. Budget 4–12 h per run
Geometries fail to mesh under sweep	High	Corner-test all 8 extremes in weeks 9–12. Narrow ranges rather than patching case by case
Publication before filing destroys novelty	Medium	Week 22 filing is a hard dependency on every submission. IPR cell engaged week 4
Scope creep to transient whole-system	Medium	Out of scope in the week 3 document. It is the Future Work paragraph
Too few usable runs	Medium	300 attempts for 250. If under 200, drop ap_t and R_bond
Workstation unavailable	Medium	Book formally for weeks 13–18 and 22–25. Split into four CSV batches

14.2 The next seven days

1. Confirm the COMSOL 6.2 licence covers CFD, Heat Transfer, Optimization and CAD Import modules. Optimization is essential to Section 9.
2. Confirm MATLAB licence covers Simulink, and whether Simscape and Stateflow are included.
3. Run the Reynolds calculation across the full range yourself. Confirm laminar. Write it up as equation one.
4. Obtain Gunjo et al. 2017 and the Sun et al. data via Sakib's reference 49.
5. Download a TMY weather file for your city from the EnergyPlus weather database.
6. Create the Fusion 360 parameter table from Section 6.1 before drawing anything.
7. Build the simplest COMSOL model — one straight channel, uniform flux, no radiation. Verify the energy balance closes to 1 % against $Q = \dot{m} \cdot c_p \cdot \Delta T$ by hand.
8. Email the IPR cell about the student patent filing process.

15. Revision H — corrections and as-built verification

This appendix is new in Revision H. It records what was corrected in the sections above, the evidence for each change, and the state of the CAD model that the corrections were verified against. Every figure quoted as as-built is a measurement taken from GRAIL_COLLECTOR_MASTER at version 12, not a target.

15.1 Correction register

#	Section	What was wrong	What Revision H does
C1	4.1 A7, 6.1, 8.2	The CAD had the channel grading diverging, Dh rising 5.2 to 8.345 mm. A7 gives G_ratio a range of 0.35 to 1.50, 6.1 states 0.54 with $Dh_{out} = Dh_{in} * G_ratio$, and 8.2 gives D 0.0028 to 0.0052 m. All three specify a converging channel.	Geometry rebuilt converging. Measured Dh ratio 0.540339 and Re 23.9 to 198.8. All earlier CFD output is void.
C2	4.1 A8	lam_G was set to 2.0 in the model on the authority of a reference image that was later shown to contradict the roadmap on the grading direction.	Returned to the A8 baseline of 1.0. Endpoints are independent of the exponent, so Dh and Re are unchanged.
C3	5.2	The TOTAL row read 17.82 kg. The line items above it sum to 22.42 kg.	TOTAL corrected to 34.13 kg, which is the corrected items plus the two additions below.
C4	5.2 and 5.3	VIP listed at 1.05 kg for 1120 x 500 x 20 mm, implying 93.75 kg/m ³ , while 5.3 gives VIP fumed silica as 190 kg/m ³ .	Mass corrected to 2.09 kg at the corrected panel size. 5.3 is right; the mass was wrong.
C5	5.2	VIP length 1120 mm overlaps the manifold blocks at x = plus and minus 550 mm.	Trimmed to 1100 mm, the absorber length. Added to 6.1 as vip_L.
C6	5.1 against 6.1	PCM thickness given as 11.6 mm in 5.1 and 10 mm in 6.1.	5.1 corrected to 10 mm, which is the value that reproduces the 4.65 kg the BOM states.
C7	5.2	Frame mass 2.90 kg for 3.6 m implies 0.806 kg/m, a 298 mm ² section, roughly a 0.9 mm wall on a 92 mm deep glazed profile.	Left OPEN. Below the practical minimum for structural 6063-T6; needs a stated wall thickness, then the mass follows.
C8	5.2	Manifold mass 0.35 kg for two blocks is 0.175 kg each, and no manifold dimensions are stated anywhere in the document.	Left OPEN. A block spanning the 480 mm absorber width with two headers and six ports cannot be that small.
C9	5.1 and 5.2	The built collector is double glazed and has mounting hardware. Revision G lists one pane and no mounting at all.	Inner glazing pane and mounting hardware added to both sections. The 15 mm cavity is now shown as an envelope divided into 7.45 and 4.30 mm.
C10	7.1 item 4	Instructs Form Union. On this geometry in COMSOL 6.2 that produced no usable exterior fluid boundary.	Item 4 rewritten for Form Assembly with an explicit check that the Continuity node exists.
D4	4.1 A11	Not an error. A6, A11 and 6.1 cannot all hold at once: at a 1.0 mm fillet the section degenerates before the 6.1 grading is reachable.	Root fillet reduced to 0.5 mm, inside the range A11 already permits. Role changed from FIXED to SWEPT.

#	Section	What was wrong	What Revision H does
C11	4.3 C5, 4.4, 8.4, 1	The second glazing pane recorded in C9 was never carried into the optics. C5 tau_glz 0.91 is a SINGLE pane value and 8.4 builds the absorber boundary condition from it, so every CFD run would over-predict absorbed flux.	Group C gains C9 tau_glz_sys, Group D gains D11 glz_n_panes, and the 8.4 boundary condition now uses tau_glz_sys. Section 1 optical penalty restated from 18 to about 25 per cent.
C12	10.2, 10.3, 1, 11.2	Numbers that depend on corrections made earlier in this revision were left behind: the ANN input width still read 15 after D4 widened the 4.8 active set to 16, and the Section 1 and 11.2 efficiency figures were derived from the single-pane optical term that C11 corrected.	Input width corrected to 16 in 10.2 and 10.3. Section 1 and 11.2 now state that the efficiency figures are single-glazed design targets, with the optical term falling from 0.66 to about 0.60, and that the loss term must be recomputed alongside it.
C13	3, 13.2, 15.9, 15.3	A full read of the sections not previously reviewed found a wrong cross-reference, and found that the Filing 1 conflict recorded in 15.9 reaches Filing 2 as well as Filing 3.	Section 3 now points to 13.8 for the search strings rather than 13.6. 15.9 extended to cover Filing 2 and to state the common cause. 15.3 gains an open manufacturing question on the 0.5 mm fillet.
C14	8.1, 9.1, 15.12	Section 8.1 specified a three-channel strip with symmetry but no such domain existed, and the symmetry boundary it specifies is not exact under alternating flow. Separately, no check of any kind had been run on the invented manifold dimensions.	The strip is built and its extents recorded in 8.1, together with an exact 2-channel periodic alternative and the reasoning for choosing between them. A manifold distribution hand check is recorded at 4.8 per cent header-to-channel pressure drop, with the failed area rule stated.

15.2 C1 — the grading direction

This is the correction that matters most, because every CFD result produced before it ran through the wrong geometry. The model had been built to a reference image supplied outside this document, showing hydraulic diameter rising from 5.2 mm to 8.3 mm. That is a Dh ratio of 1.596, which is outside the range A7 states, contradicts the value 6.1 gives, and breaks the Reynolds band of 8.2.

A rising hydraulic diameter lowers Reynolds number, so the diverging model ran at Re 15 to 107 where 8.2 states 30 to 210. Checking the built geometry against 8.2 would have caught this immediately. That check is now mandatory and appears in 15.8 below.

Quantity	Before	After	Roadmap
Hydraulic diameter at inlet	5.20135 mm	5.199685 mm	5.2000 mm, A6
Hydraulic diameter at outlet	8.34506 mm	2.809592 mm	2.8080 mm, 6.1
Dh ratio	1.596	0.540339	0.54, A7 and 6.1
Reynolds across the mdot range	15 to 107	23.9 to 198.8	30 to 210, 8.2
Section scale at outlet	1.629546	0.527853	derived
Lateral conduction bridge	26.195 mm	31.299 mm	20 to 60 mm, B3

Note that w_bridge moved closer to the B3 baseline of 34.8 mm once the grading was corrected, from 26.2 to 31.3 mm. With a converging channel the widest section sits at the inlet, which is what sets the minimum bridge. That is independent corroboration that converging was the intended direction. The driven expression

must therefore be $w_{\text{bridge}} = ch_{\text{pitch}} - ch_{\text{w_in}} * \max(1, G_{\text{ratio}})$, so that it remains the true minimum under a sweep that crosses $G_{\text{ratio}} = 1$.

15.3 Decision D4 — the root fillet

The channel section is a flat bond land, a root fillet and a circular dome, all tangent, so the dome radius follows from the other three: $R = ((h - r_f)^2 + ((w - 2*r_f)/2)^2 - (r_f + t)^2) / (2*((h - r_f) - (r_f + t)))$, where h and w are the section height and width, r_f the root fillet and t the roll-bond sheet thickness.

The denominator passes through zero when $h = 2*r_f + t$ and goes negative below it, which inverts the dome into a concave trough. At a section scale of 0.62 with the 1.0 mm fillet the sketch solved with a dome radius of minus 0.8703 mm and the CAD package reported only a warning, not an error. An inverted channel can pass unnoticed and mesh successfully, so any sweep must assert that the computed dome radius is positive rather than relying on the tool to object.

Root fillet	Degeneracy at $h =$	Minimum section scale	Lowest reachable Dh ratio
1.0 mm, the A11 baseline	3.000 mm	0.745525	about 0.776, so 0.54 is impossible
0.5 mm, the A11 lower bound	2.000 mm	0.483915	0.54 reachable with margin

Measured on the model at the 1.0 mm fillet, approaching the floor flattens the dome toward a plate rather than narrowing the channel: section scale 0.80 gave Dh_{out} 4.227 mm at a dome radius of 15.1 mm, 0.78 gave 4.131 mm at 21.2 mm, and 0.76 gave 4.035 mm at 44.3 mm. Reynolds topped out near 135 against the 210 that 8.2 states.

Decision D4 therefore reduces r_{fillet} to 0.5 mm. It was the only option that makes A7, 6.1 and 8.2 simultaneously true. Two consequences must be carried forward: A11 lists the Role of r_{fillet} as FIXED, so moving it is a documented deviation and is why the Role now reads SWEPT; and 6.3 notes that the fillet is where the lateral conduction path lives, so F10 must sweep r_{fillet} alongside w_{bridge} rather than treating it as frozen.

Changing the fillet changes the inlet section, so $ch_{\text{h_in}}$ was re-solved to hold A6 exactly. The 8.0 by 3.8 mm section that 6.1 gave does not produce $Dh_{\text{in}} = 5.2$ mm with any fillet in the A11 range. A6 is the binding statement and the section follows from it: $ch_{\text{h_in}}$ 4.132953 mm, $ch_{\text{w_in}}$ 8.700954 mm, aspect ratio 2.105263 against the A10 target of 2.1, and measured Dh_{in} 5.199685 mm.

One question Decision D4 raises and does not answer. Section 9.1 gives the roll-bond manufacturable minimum feature size as about 3 mm, and dependent claim 5 of Filing 1 recites a minimum feature size not less than 3 mm. Those refer to the width of a flow-field feature, not to a blend radius at a channel root, so there is no contradiction with a 0.5 mm fillet — and A11 has always permitted 0.5 mm. But D4 has now placed r_{fillet} at the extreme of that range, and whether the roll-bond inflation process can actually hold a 0.5 mm root radius is a manufacturing question that nobody has put to a supplier. Add it to verification item V6 when the TIM, VIP, PCM and coating datasheets are requested. If the process cannot hold it, the fillet floor rises again and the grading of 6.1 becomes unreachable a second time — so this is worth asking early rather than late.

A SECOND OPEN QUESTION, added in Revision I and belonging with the one above. Every manifold dimension in the model is an ENGINEERING_ASSUMPTION — header bore, header length, riser and port bores, block heights, port position. None came from a supplier and none is checked by CFD, because Section 8.1 excludes the manifold from the domain. A hand check was run instead, at the E4 design point with water at ρ 984 and μ 4.9e-4: header bore 9 mm carrying six channels gives 0.0200 m/s and Re 361, against channel velocities of 0.0075 m/s at inlet and 0.0267 m/s at outlet. Laminar pressure drop comes to 0.93 Pa along the 480 mm header against 19.43 Pa along the 1100 mm channel, a ratio of 4.8 per cent — inside the 10 per cent rule of thumb, so channel-to-channel maldistribution should be a few per cent. Note what fails: the header bore area of 63.617 mm² is only 0.378 times the 168.331 mm² combined inlet area of the six

channels it feeds, which breaks the usual area rule outright. It passes on pressure drop only because the channels are long and narrow enough to dominate the resistance. That is a thin margin resting on one hand calculation. Put the header bore to the supplier with the fillet question at V6, and if the manifold is ever suspected in test, model it explicitly rather than trusting this paragraph.

15.4 C2 — the grading exponent

A8 gives lam_G a baseline of 1.0 and a swept range of 0.5 to 3.0. The CAD carried 2.0, justified in its own parameter note as following the reference image label Exponential. Once that image was shown to contradict the roadmap on the grading direction it ceased to be a valid source for the exponent either, so lam_G returns to the A8 baseline and the exponent is left to F10 to sweep rather than asserted by the model.

This costs nothing geometrically. The grading law is $Dh(x) = Dh_{in} * [1 - (1 - G_{ratio}) * (x/ap_L)^{lam_G}]$, and at $x = ap_L$ the bracket equals G_{ratio} for any exponent, so the endpoints are independent of lam_G. Only the interior profile and the fluid volume change.

15.5 Two withdrawn claims

An earlier working note flagged that 40 minus 5.2 equals 34.8 exactly, and inferred that B3 had been computed as pitch minus hydraulic diameter, using a diameter where a width belongs. That inference is withdrawn. B3 is a swept parameter with a stated range of 20 to 60 mm, the built value of 31.299 mm falls inside it, and an arithmetic coincidence is not evidence of a mistake.

A second note instructed a future reader to correct 8.2 before quoting a Reynolds range. That was backwards and is also withdrawn. Section 8.2 is correct: its D range of 0.0028 to 0.0052 m encodes the converging channel, and 0.0028 divided by 0.0052 is 0.538, the same 0.54 that 6.1 states. The model was wrong, not the roadmap.

15.6 Note on section 5.3

Every density in 5.3 was checked against the CAD by dividing measured mass by measured volume for all 138 bodies. All fifteen materials match to the last digit printed, and 5.3 is unchanged in this revision. A working note that claimed 5.3 gave PC honeycomb as 851 kg/m3 was a transcription error in the note; 5.3 reads 85, the model uses 85, and the resulting 0.456 kg agrees with the 0.42 kg the BOM states.

One point worth stating explicitly: 5.3 gives the TiNOX coating no density, because the row is a boundary condition, and 5.2 gives it no mass. That is correct, and 8.3 handles the coating as a thin layer rather than a meshed solid. It does mean a CAD package that insists on a density for a solid body has nothing to take from this document. The model assumes 4500 kg/m3, which is typical for TiNOx on aluminium, and with the coating built at its true 0.3 micron thickness this contributes 0.00078 kg. That assumed density must never be quoted as a roadmap value.

15.7 As-built verification record

GRAIL_COLLECTOR_MASTER, Fusion 360, parametric, version 12. Nineteen verification gates, all passing.

Check	Measured	Roadmap	Result
Inlet hydraulic diameter	5.199685 mm	5.2000, A6	PASS
Outlet hydraulic diameter	2.809592 mm	2.8080, 6.1	PASS
Dh ratio	0.540339	0.54, A7 and 6.1	PASS
Reynolds, full mdot range	23.9 to 198.8	30 to 210, 8.2	PASS

Check	Measured	Roadmap	Result
Grading exponent	1.000	1.0, A8	PASS
Root fillet	0.500 mm	0.5 to 2.0, A11	PASS
Channel aspect ratio	2.105263	2.1, A10	PASS
Lateral conduction bridge	31.299 mm	20 to 60, B3	PASS
Channel count and pitch	12 at exactly 40.00 mm	12, ap_W/n_ch, A4 and A5	PASS
Channel volume spread	0.000000000 mm3	identical by symmetry	PASS

Model integrity check	Result
Timeline	945 features, 0 unhealthy, 1 deliberately suppressed
Bodies	138, all solid and valid, 0 default-named, 0 stray at root
Materials	15 materials, every density matching 5.3 exactly
Interference	476 candidate pairs from 9,453, tested by boolean intersection, 0 real interferences
Mounting symmetry	mirror-checked body by body in X and Y, 0 mismatches
CFD export duplicates	fluid circuits match the 12 channel voids plus 4 manifold voids to 0.0000 mm3
Parameters	115, all evaluating, all carrying a provenance comment
Open items in the CAD	0
Hardware dry mass	43.456 kg, with 0.445 kg of water and sealed air

The alternating counter-current arrangement that Filing 1 claims was verified by boolean intersection against the two circuit bodies rather than by inspection. The channel circuit sequence in Y order is A B A B A B A B A B A B. The odd circuit runs from the left cold header to the right hot header and the even circuit from the right cold header to the left hot header. Twelve ports, six per end, cold at $z = -11$ mm and hot at $z = +14$ mm, mirror-symmetric left to right.

15.8 What remains open, and the corrected campaign sequence

Nothing remains open in the CAD. Two items remain open in this document, C7 and C8, because closing them requires a decision about actual manufactured dimensions rather than a correction. The frame needs a stated wall thickness, from which its mass follows; the manifolds need stated dimensions, without which every manifold dimension in the CAD remains an engineering assumption.

1. Re-export the STEP. Suppress the 57 components the CAD lists as CFD_suppress_count, being 30 fasteners, 5 seals, 14 mounts and 8 sensors, plus the selective coating, 58 bodies in all. Do not suppress the 12 ports, which carry the inlet and outlet faces.
2. Import per the corrected 7.1: Form Assembly, confirm the Continuity node exists, then run the trivial isothermal check that 7.1 already requires, mass out equal to mass in within 0.1 percent and the fluid domain connected end to end.
3. Confirm that Reynolds falls in 23.9 to 198.8 before anything else is run. This is the cross-check whose absence allowed C1 to survive as long as it did.
4. Close the validation gate of 8.7 before building any variant: Sun et al. under 3 percent, Gunjo et al. under 5 percent, hard stop at 8 percent, debugging in the order 8.7 gives, pure conduction, then adiabatic flow-only, then radiation.

5. Only then build variants. Run the counter-current comparison at matched mean plate temperature, and report the matched-conditions result alongside it.

6. Sweep w_{bridge} across the full 20 to 60 mm that B3 states, and per Decision D4 sweep r_{fillet} 0.5 to 2.0 mm with it, because the fillet now carries part of the lateral conduction path the claim depends on.

One standing caution for the thesis and the filings. Every performance number recited in a claim must be demonstrated by the corrected model and not inherited from the earlier one. All CFD output produced before correction C1 was computed on a channel that widened where it should have narrowed; it is void rather than approximately right, and none of it may be quoted. Filing 1 claims a conductive bridge width selected to reduce the peak-to-mean temperature difference of the solar-facing surface. That is a claim about the temperature field, not about efficiency, so keep the claim and the evidence aligned: demonstrate the flatter field, and report efficiency separately and honestly.

15.9 Cross-check against the draft claims of 13.6

The draft claims were checked against the built model. This is an engineering cross-check, not legal advice; V10 still requires review by a registered patent agent.

Dependent claim 1 of Filing 1 recites that hydraulic diameter DECREASES MONOTONICALLY along the flow direction, with G between 0.35 and 1.00. That is a fourth independent statement in this document that the channel converges, alongside A7, 6.1 and 8.2, and it is the most consequential of the four. The pre-correction geometry had a D_h ratio of 1.596, which is outside the range the claim recites, so no result computed on that geometry could have supported Filing 1 at all. The corrected model measures 0.540339 and sits inside the claim.

A discrepancy to resolve before filing: A7 gives G a swept range of 0.35 to 1.50, but dependent claim 1 caps it at 1.00. Any sweep point above 1.00 is a diverging channel that the claim does not cover. Such points may be studied, but they cannot be recited as embodiments, and a result drawn from one must not be offered in support of the claim. Either narrow the A7 sweep to 1.00 for claim-supporting runs, or widen the claim and accept the prior-art exposure that a diverging channel brings.

Dependent claim 3 recites a conductive bridge between 20 and 60 mm. As built it is 31.299 mm, inside the range, and closer to the B3 baseline of 34.8 mm than the pre-correction 26.195 mm was. Dependent claim 4 recites a transparent insulation layer below 0.10 W/m.K through-thickness; 5.3 gives PC honeycomb 0.075 perpendicular, which satisfies it.

Filing 3 needs attention, and this is the one substantive conflict the cross-check found. Its independent claim recites that the areal mass of phase change material increases monotonically IN THE DIRECTION OF WORKING FLUID FLOW, reaching at least 1.5 times the inlet value. The built PCM does grade monotonically, from 7.143 mm at $x = -550$ to 12.857 mm at $x = +550$, a ratio of 1.800, which satisfies the 1.5 times requirement. But under the alternating counter-current arrangement that Filing 1 claims there is no single direction of working fluid flow: the odd channels run from $x = -550$ to $x = +550$ and the even channels run the opposite way. The PCM therefore thickens toward the outlet for half the channels and toward the inlet for the other half.

As drafted, Filing 1 and Filing 3 describe the same collector but assume incompatible temperature fields. Filing 1 exists precisely because alternating the flow FLATTENS the axial temperature rise and puts hot and cold regions in lateral adjacency; Filing 3 assumes a monotonic axial rise so that concentrating PCM near the outlet is worthwhile. Three ways out, for the patent agent to weigh: recite the PCM grading against the local time-averaged absorber surface temperature rather than against flow direction, which is what the physics actually motivates; or restrict Filing 3 to the parallel-flow embodiment and file it independently of Filing 1; or grade the PCM laterally to match the alternating pattern, which would be a different and possibly stronger claim. Do not file the two as drafted without resolving this.

The same objection reaches Filing 2, and this was missed when 15.9 was first written. Section 13.2 describes its relief zone as sitting "adjacent the outlet where the plate runs hottest", and the independent claim in 13.6 recites that the second region "is located adjacent the working-fluid outlet". Under the alternating counter-current arrangement there is no single outlet end: half the channels discharge at each end of the panel, so a relief zone placed at one end sits adjacent the outlets of half the channels and the inlets of the other half.

Taken together this is one problem, not three. Filing 1 works by FLATTENING the absorber temperature field — that is its entire technical effect, and the reason it reduces radiative loss at unchanged mean temperature. Filings 2 and 3 both exploit a PEAK in that same field: Filing 2 puts a radiative relief zone where the plate is hottest, Filing 3 puts latent capacity there. The better Filing 1 works, the less peak remains for the other two to act on. They are not merely worded for a different embodiment; they are in physical tension with the mechanism Filing 1 claims.

This does not sink any of the three, and it may even sharpen them. Filing 1 stands entirely on its own. Filings 2 and 3 need their claims recited against the local time-averaged absorber surface temperature rather than against flow direction or outlet position, which is what the physics actually motivates and which remains true whichever flow arrangement is used; or they need to be filed against the parallel-flow embodiment and kept clear of Filing 1. What must not happen is the three being filed as drafted, describing one collector on three incompatible assumptions about its temperature field. Raise this with the patent agent at the Week 20 drafting meeting, and note that Section 13.10 item 3 already schedules the counter-current and graded-PCM simulations in the same campaign — the data needed to settle it will exist before the drafting date.

15.10 C11 — the optical cost of the second glazing pane

Correction C9 added the inner glazing pane to the layer stack and the bill of materials, because the built collector has one. What C9 did not do, and what this correction repairs, is carry that pane into the optics. It is the most consequential omission found after C1, because it does not merely describe the collector wrongly — it would have inflated every efficiency number the project produces.

C5 gives τ_{glz} as 0.91 with a range of 0.88 to 0.92. That is a SINGLE pane value. Section 8.4 then builds the absorber boundary condition as $q = G_T \text{ times } \tau_{glz} \text{ times } \tau_{tim} \text{ times } \alpha_{abs} \text{ times } (1 \text{ minus } f_{shade})$, taking that single-pane figure. With two panes in the built stack the glazing system transmits materially less, and no value in the C5 range can represent it.

First-order estimate for two identical low-iron panes, each τ 0.91 with reflectance about 0.08: $\tau_{sys} = \tau \text{ squared divided by } (1 \text{ minus } \rho \text{ squared}) = 0.8281 / 0.9936 = \text{about } 0.833$. At the design point this takes absorbed flux from $800 \times 0.91 \times 0.82 \times 0.95 = 567.1 \text{ W/m}^2$ down to $800 \times 0.833 \times 0.82 \times 0.95 = 519.1 \text{ W/m}^2$, a loss of 48 W/m^2 or 8.5 per cent. The optical penalty that Section 1 quotes as 18 per cent becomes about 25 per cent.

Treat 0.833 as an engineering estimate and nothing more. It ignores angular dependence, the spectral mismatch between the two panes and the absorber, and the AR and TiO₂ coatings that 5.1 places on the outer pane. Compute τ_{sys} properly in the optical model and validate it before any efficiency figure is quoted. The new Group C row C9 carries it as DERIVED for exactly that reason.

A DESIGN QUESTION THIS OPENS, which is not ours to close. The second pane costs about 8.5 per cent of absorbed flux and buys a reduction in U_L by splitting the 15 mm cavity into two of 7.45 and 4.30 mm. Whether that trade is favourable at the 40 K operating difference this collector targets has not been computed, and the answer decides whether the pane stays. Run the comparison — single pane against double, at matched U_L method and matched mass flow — before the design is frozen. If the double-glazed stack loses, drop the pane and revert C9 rather than carrying a penalty the design does not earn.

15.11 C12 — numbers that depended on the corrections above

Two corrections made earlier in this revision had consequences elsewhere in the document that were not followed through at the time. Both are repaired here, and both are worth recording because they are the same failure mode that produced C1: a number changed in one place and the places that depended on it were not revisited.

First, the surrogate input width. Decision D4 moved r_{fillet} from FIXED to SWEPT, which widened the 4.8 active set from fifteen parameters to sixteen. Section 10.2 still specified fifteen input columns and 10.3 still specified a 15 to 64 to 32 to 16 to 1 architecture. Built as written, the network would have silently dropped a swept parameter — and it would have dropped the very one that Decision D4 added because it shares the lateral conduction path with w_{bridge} . Both are now sixteen.

Second, and more consequential, the efficiency figures. The Section 1 headline table derives $F_R(\tau \cdot \alpha) = 0.66$ explicitly from $\tau_{\text{glz}} 0.91$ times $\tau_{\text{TIM}} 0.82$ times $\alpha 0.95$, and 11.2 carries the same 0.66 into the expected quadratic coefficients. Correction C11 established that the built collector is double glazed and that the correct optical term is $\tau_{\text{glz_sys}} 0.833$. That alone takes $F_R(\tau \cdot \alpha)$ to about 0.60.

The efficiency rows have deliberately NOT been recomputed. Correcting only the optical term would overstate the harm, because the same second pane that costs transmittance also lowers U_L by splitting the cavity, and the $F_R \cdot U_L$ figure of 2.00 was itself computed for the single-pane stack. Recomputing one term without the other would produce a number that is precise and wrong. Both sections now carry an explicit warning that their efficiency figures are single-glazed design targets, which is what they have always been.

For scale, and as an illustration rather than a result: correcting the optical term alone moves the crossover against a conventional collector from about 20 K to about 30 K, and the headline gain at 40 K from +27 per cent to roughly +14 per cent. Crediting the U_L improvement moves both back. The honest position is that the net is undetermined until the single-pane against double-pane comparison in 15.10 is run, and that no efficiency number from this document should be quoted until it is.

15.12 C14 — the CFD domain, and a boundary condition that was wrong as specified

Section 8.1 has always carried the right instinct and the wrong boundary condition. The instinct is that the alternating counter-current arrangement cannot be studied on a single channel, because the effect is lateral and a single channel has no neighbour — on that domain Filing 1 measures exactly zero, not a small number. The wrong part is the two words "with symmetry".

A symmetry plane sets lateral heat flux to zero and is valid only where the temperature field mirrors about it. Cut at a bridge midline under alternating flow and the neighbour across the cut runs the opposite direction: the field mirrors in y only if it is also reversed in x . That is a glide symmetry, and an adiabatic plane is not the right condition for it. The error sits precisely where the mechanism under study does its work, which is the worst place for it.

Both domains are now built so the choice can be made on evidence rather than on which one exists. The three-channel strip `CFD_STRIP_3CH_MASTER` keeps Section 8.1 as written and pays for the approximation with a third channel: with channels at $y = -60, -20$ and $+20$ and cut planes at -80 and $+40$, the middle channel is a full channel and a full bridge away from either boundary, so results read from it are interior results. The two-channel domain `CFD_STRIP_2CH_PERIODIC` spans y from -80 to 0 with translational periodicity of period 80 mm, which is not an approximation at all — it is the exact statement that the panel repeats every two channels. Its periodic faces measure 2200.000000 mm² each and are identical rectangles in the bridge material, so they pair cleanly.

The periodic domain is both more correct and about a third cheaper, which is an unusual combination and worth acting on. What must not happen is results from the two domains being mixed into one surrogate

training set, or a claim in the thesis resting on the symmetry domain without the caveat in this section attached to it. Make the choice, write it down, and run the whole sweep on one domain.

One thing this correction does not change. The pre-correction geometry problem that C1 fixed was a wrong channel; this is a wrong boundary. Both have the same shape as a failure — a specification that was carried forward for months because nobody built the thing it described and checked it. The strip now exists, its extents are in 8.1 to six decimal places, and the conjugate interface has been verified point by point. The document and the model agree on this section for the first time.

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